



AI Requirements Framework

for Safety-Critical Systems

Requirements and Verification Standards for Artificial Intelligence in Safety-Critical Applications

V2.1.2

Safety Critical Labs
AI CERTIFICATION AUTHORITY

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Section 1: Introduction

1.1 PURPOSE

This framework establishes the AI-specific requirements and verifications against which Safety Critical Labs conducts certification assessments for systems incorporating artificial intelligence or machine learning capabilities in safety-critical applications.

This framework:

- Establishes requirements addressing AI and ML-specific failure modes not covered by traditional software verification practices
- Defines verification methods and evidence standards for data-driven, probabilistic systems
- Provides AI-specific requirements for bias, drift, hallucination, explainability, and human-AI interaction
- Applies across aerospace, aviation, automotive, medical, and other safety-critical domains

Rationale: *Traditional software assurance practices assume deterministic, logic-based systems with stable behavior after deployment. AI and ML systems challenge this paradigm through probabilistic outputs, emergent behavior from training data, and potential performance degradation over time. These characteristics require verification approaches designed specifically for data-driven systems.*

1.2 SCOPE AND APPLICABILITY

This framework applies to any system, subsystem, or component incorporating artificial intelligence or machine learning capabilities that impacts safety-critical or mission-critical operations. The framework is domain-agnostic and applicable to aerospace, automotive, medical, industrial, and other safety-critical domains.

Table 1-1: Applicability Determination Process

Step	Action	Outcome
1	Identify AI/ML presence using Section 1.2.1 criteria	Yes → Step 2; No → standard software assurance practices apply
2	Classify AI system (Safety-Critical, Mission-Critical, Operational Support)	Determines applicable requirements

Step	Action	Outcome
3	Document applicability determination	Retained as verification evidence
4	Establish tailoring if applicable	Tailoring rationale documented

Should tailoring of listed requirements be considered, reference Section 1.2.6 on how to assess tailoring approaches.

1.2.1 Defining AI

Traditional software behavior is explicitly defined by code. Verification confirms the code implements the required logic, and once verified, behavior remains stable unless the code is changed. AI/ML systems operate differently behavior emerges from patterns learned during training rather than from explicitly programmed instructions.

This fundamental difference means code inspection alone cannot verify AI behavior, validated performance may not persist over time, and outputs may be non-deterministic. These characteristics do not make AI systems unsuitable for safety-critical operations, but they do require verification approaches designed for data-driven, emergent behavior. The requirements in this framework provide those approaches.

Table 1-2: Applicability Criteria

Criterion	Indicator	Verification Impact
Learned behavior	System uses behavior derived from training data rather than explicitly programmed logic	Behavior cannot be verified by code inspection alone
Probabilistic outputs	System outputs are non-deterministic or include confidence/uncertainty	Identical inputs may not produce identical outputs
Potential drift	System behavior may change or degrade over time without code modifications	Validated performance may not persist without ongoing monitoring
Decision impact	System makes or informs decisions affecting human safety or operational success	Undetected failures may have severe consequences

A system meeting one or more criteria: proceed to Section 1.2.2 to determine classification and applicable requirements. A system meeting no criteria: this framework does not apply; use standard software assurance practices.

1.2.2 AI System Classification

Once a system has been identified as meeting the applicability criteria in Section 1.2.1, it shall be classified according to Table 1-3. Classification determines which requirements apply and the level of tailoring authority available.

Table 1-3: AI System Classification

Classification	Description	Applicable Requirements
Safety-Critical AI	AI outputs directly affect human safety or system survivability	All AI-1 through AI-10
Mission-Critical AI	AI outputs affect operational success but not human safety	AI-1 through AI-6, AI-8, AI-9, AI-10; AI-7 tailored
Operational Support AI	AI supports operations but does not drive critical decisions	AI-1 through AI-6 and AI-10 minimum

Systems meeting the Adaptive learning criterion in Table 1-2 shall be classified no lower than Mission-Critical and require tailoring approval per Section 1.2.6.

Classification also triggers the ODD declaration rigor specified in Section 1.2.3 Table 1-4, and AI-10 (Privacy and Data Protection) applies where the system processes personal data under any applicable privacy framework; where personal data is not processed, AI-10 sub-requirements may be documented as Not Applicable with rationale.

1.2.3 Operational Design Domain

Following the procedural pattern of Section 1.2.1 (Defining AI) and Section 1.2.2 (AI System Classification), this section establishes ODD declaration as a process step that runs alongside classification. The formal requirement is established in AI-1.0.

The AI system shall have a declared Operational Design Domain (ODD) prior to verification. The ODD defines the operational conditions, inputs, users, subjects, regulatory context, and constraints under which the system is designed and validated to operate, and serves as the authoritative scope statement for the certification.

Declaration rigor scales with classification per Table 1-4. The formal requirement and verification for ODD declaration are established in AI-1.0. Tailoring of ODD declaration follows the tailoring authority defined in Section 1.2.6.

Table 1-4: ODD Declaration Rigor by Classification

Classification	Required Declaration Rigor
Safety-Critical AI	Operational Claim summary statement plus structured ODD using the full attribute taxonomy (all eight categories) with quantitative bounds for all measurable attributes. Exclusions explicitly enumerated using all five exclusion sub-categories. ODD declaration formally reviewed and approved by the Project Safety Review Board.
Mission-Critical AI	Operational Claim summary statement plus structured ODD addressing all eight attribute categories. Quantitative bounds where measurable, qualitative descriptors where appropriate. Exclusions explicitly enumerated using applicable sub-categories. ODD declaration approved by the Chief Engineer.
Operational Support AI	Operational Claim summary statement plus documented ODD addressing Operational Environment, Operational Phases, Regulatory and Operational Authorization, Subject Population (where applicable), and Exclusions at minimum. Other categories addressed as applicable to the AI function. ODD declaration approved by the Software Assurance authority.

1.2.4 Relationship to Traditional Software Standards

Traditional software engineering standards were developed for deterministic, logic-based systems. These standards implicitly assume characteristics that do not hold for AI/ML systems. This section documents the gaps that necessitate AI-specific requirements.

1.2.4.1 Gap Analysis

Table 1–5: Traditional Software vs. AI/ML Gap Analysis

Traditional Software	AI/ML Reality	Risk If Unaddressed
Fault = bug in code	Fault = bias, drift, hallucination	Wrong failure mode analysis
V&V at delivery	Requires continuous validation	Post-deployment failures undetected
Requirements to Code traceability	Behavior emerges from training data	Loss of traceability
Deterministic outputs	Probabilistic outputs	Unpredictable behavior in edge cases

Traditional Software	AI/ML Reality	Risk If Unaddressed
Code coverage metrics	No equivalent for trained models	Inadequate test coverage
Explicit decision logic	Black-box decision making	Operators cannot verify AI reasoning
Scope implicit from documentation	Validated envelope varies by training data	Unbounded certification claim without explicit operational scope
Privacy handled procedurally outside the system	Personal data can be reconstructed from models, membership inferred, decisions evade data subject rights	Privacy regulation violation; model inversion and membership inference risk

1.2.4.2 Gap-to-Requirement Mapping

Each AI-specific requirement set in Section 2 addresses one or more gaps identified above. Table 1-6 provides traceability from identified gaps to the corresponding requirements.

Table 1-6: Gap-to-Requirement Mapping

Gap / Risk	Requirement Set
Unbounded operational scope	AI-1.0: Operational Design Domain
Training data integrity and traceability	AI-1: Operational and Data Foundations
Bias and fairness issues	AI-2: Addressing AI Bias
No test coverage equivalent	AI-3: ML Test Coverage
Static after deployment / drift	AI-4: Continuous Validation
Unpredictable outputs	AI-5: Hallucination Prevention
Vulnerable to OOD inputs	AI-6: Out-of-Distribution Detection
Vulnerable to adversarial attacks	AI-7: Adversarial Robustness
Black-box decision making	AI-8: Explainability
Operators cannot verify reasoning	AI-9: Human-AI Teaming
Personal data protection and data subject rights	AI-10: Privacy and Data Protection

1.2.5 Integration with Software Lifecycle

All AI software shall also meet existing software engineering requirements. AI systems are software systems, and the foundational practices of software planning, configuration management, verification, and safety assurance remain applicable. This framework does not create a parallel lifecycle; rather, it identifies AI-specific activities that shall be performed within each phase of the existing software lifecycle.

Table 1–7: AI Integration with Software Lifecycle

Lifecycle Phase	Title	Key Artifacts
Requirements	Define AI function criticality; identify protected classes	AI applicability determination
Design	Define model architecture; establish explainability approach	AI design documentation
Implementation	Partition datasets; document provenance	Dataset partition records
Test	Execute AI-specific test matrix	Test reports, coverage reports
Verification	Complete verification Matrix; compile evidence	Verification matrix, evidence package

1.2.6 Tailoring Guidance

Should tailoring of listed requirements be considered, Table 1–8 identifies the tailoring authority for each AI system classification. All tailoring shall be documented with rationale and retained as verification evidence. Changes to ODD declarations (AI-1.O) follow the same tailoring authority as other requirements in the corresponding classification.

Table 1–8: Tailoring Authority by Classification

Classification	Tailoring Authority
Safety–Critical AI	No tailoring without Project Safety Review Board approval
Mission–Critical AI	Tailoring permitted with Chief Engineer approval and documented rationale
Operational Support AI	Tailoring permitted with Software Assurance authority approval

1.3 NORMATIVE REFERENCES

The following documents are referenced by this framework. Applicable standards are cited within individual requirements and verification methods.

Table 1-9: Applicable Standards

Document	Title	Relevance
ISO/IEC 22989:2022	AI Concepts and Terminology	Definitions
ISO/IEC 5338:2023	AI System Lifecycle Processes	Lifecycle
NIST AI RMF 1.0	AI Risk Management Framework	Risk taxonomy
DO-178C	Software Considerations in Airborne Systems	Aviation software
ISO 26262	Road Vehicles – Functional Safety	Automotive safety
ISO/IEC 27701:2019	Privacy Information Management	Privacy controls
ISO/IEC 29100:2011	Privacy Framework	Privacy principles
NIST Privacy Framework v1.0	Enterprise Privacy Risk Management	Privacy risk taxonomy
EU AI Act	Regulation on Artificial Intelligence	AI risk tiering (EU)

Table 1-10: Reference Documents

Document	Title	Relevance
NPR 7150.2D	NASA Software Engineering Requirements	Software lifecycle reference
NASA-STD-8739.8	Software Assurance and Safety Standard	Safety analysis reference
ISO/IEC 23053:2022	Framework for AI Systems Using ML	ML framework concepts
ISO/IEC TR 24027:2021	Bias in AI Systems and Decision Making	Bias assessment guidance
NIST AI 600-1	AI RMF: Generative AI Profile	Generative AI risks
FAA AI Safety Roadmap	Roadmap for AI Safety Assurance	Aviation AI certification concepts
GDPR (Regulation 2016/679)	EU General Data Protection Regulation	Personal data processing (EU)
HIPAA Privacy and Security Rules	45 CFR 164 Subparts C and E	Protected health information (US)

Document	Title	Relevance
CCPA/CPRA	Cal. Civ. Code §§ 1798.100 et seq.	Consumer privacy (California)
NIST SP 800-122	Guide to Protecting the Confidentiality of PII	PII handling guidance
ISO 34503	Road Vehicles — Taxonomy for ODD	ODD attribute taxonomy (automotive)

1.3.1 Domain Standards Citation Convention

The framework requirements in Section 2 are algorithm-agnostic and domain-agnostic. However, Safety Critical Labs certification assessments are frequently conducted in the context of domain-specific regulatory regimes that impose analogous or complementary requirements. To support regulatory harmonization and reduce duplicative compliance effort for clients, each requirement cites the applicable domain standards inline at the point of reading.

Domain standards citations are organized in an “Applicable Domain Standards” subsection at the end of each parent requirement, structured by domain:

- **General:** Domain-agnostic standards (ISO/IEC, NIST, IEEE)
- **Aviation:** FAA regulations, advisory circulars, and guidance
- **Automotive:** SAE standards, ISO automotive safety standards, NHTSA guidance, UNECE regulations
- **Medical:** FDA guidance documents and regulations, HIPAA where applicable
- **EU:** EU AI Act articles, GDPR articles, and related Union regulations
- **Cross-Domain:** NIST AI RMF function and category references, NIST SP 800 series

A domain block is included only where the requirement has relevant citations in that domain. Absence of a domain block indicates that no specific domain standard currently addresses the requirement at the cited level of detail; this does not imply the requirement is inapplicable in that domain.

Domain standards cited are informative, not normative. SCL certification is conducted against the requirements in Section 2; inline domain citations support cross-regulatory recognition and facilitate compliance demonstration where a single system must satisfy multiple regulatory regimes.

In v2.1.1, anchor citations are present on all ten parent [R.AI-X] statements. Full sub-requirement-level citations appear on AI-9.1, AI-8.3, AI-2.1 (v2.1 retrofit exemplars), on the new AI-8.5, AI-9.7, AI-9.8, AI-9.9 (v2.1.1 Gap 6 and Gap 10 additions), and throughout AI-10. Applicable Domain Standards retrofit for the approximately 53 remaining sub-requirements is scheduled as a Phase 2 pass within v2.1.1.

1.4 VERB APPLICATION AND LEXICON

Requirements are defined in Section 2 and are denoted by “shall” in the requirements statement. Associated verifications are defined in Section 3.

Best practices are denoted “should” and documented in Section 4. “Should” statements do not carry a formal verification requirement.

AI-specific terminology is defined in Appendix A: Glossary, aligned with ISO/IEC 22989:2022 and ISO/IEC 5338:2023 where applicable.

1.5 THRESHOLD CATEGORY DEFINITIONS

The following threshold categories shall be documented for all AI systems regardless of classification. These categories are referenced throughout the requirements in Section 2. Threshold values are project-defined; Section 4.1 provides example ranges and selection guidance.

- **Performance degradation threshold:** Minimum acceptable accuracy, precision, recall, or domain-appropriate metric below which system response is required
- **Confidence threshold:** Minimum output confidence level for autonomous action versus human review escalation
- **Data drift threshold:** Maximum deviation from training data distribution (using KL divergence, PSI, or equivalent metric) before revalidation is triggered
- **OOD detection threshold:** Boundary conditions that classify inputs as out-of-distribution requiring fallback behavior
- **Human escalation threshold:** Conditions under which AI outputs require mandatory human review before action

Rationale: While threshold values are project-defined, these categories are necessary for adequate AI system performance and are required as evidence during certification assessment. Additional threshold categories may be needed based on specific AI function characteristics and operational context. Threshold

values shall be derived from and bounded by the declared Operational Design Domain (AI-1.0); thresholds that admit conditions outside the declared ODD are inconsistent with the certification claim.

Section 2: Requirements

The requirements in this section apply to AI systems as defined in Section 1.2.1. Each requirement set (AI-1 through AI-10) addresses a specific AI failure mode identified in Section 1.2.4. Threshold categories referenced throughout these requirements are defined in Section 1.5. Requirements are algorithm-agnostic and do not prescribe specific implementation approaches. Multi-model and neural network-specific requirements are not established in this section; instead, informative guidance is provided in Section 4.

The “will” statements are expected to be met by the sum of the supporting “shall” statements.

Note: Verifications are co-located with requirements in Section 2 for ease of review. They are summarized in the verification matrix in Section 3.

2.1 AI-1: OPERATIONAL AND DATA FOUNDATIONS

[R.AI-1] OPERATIONAL AND DATA FOUNDATIONS

The AI system will establish operational scope and data integrity sufficient to support reliable model behavior and a defensible certification claim throughout the model lifecycle.

Traditional software has direct traceability from requirements to code; an engineer can inspect the code to verify it implements the requirement. AI systems break this traceability because behavior emerges from training data within a declared operational envelope. AI-1 establishes the foundations on which all downstream verification depends: the operational envelope within which the system is designed to function (AI-1.0), the integrity of the data partitions that produced the model (AI-1.1 through AI-1.4), and the classification handling controls that govern the data and outputs throughout the lifecycle (AI-1.5 through AI-1.8).

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §5; ISO/IEC 5338:2023 §6.4
- Aviation: DO-178C §5 (Software Development); DO-330 §5 (Tool Qualification); FAA AI Safety Roadmap

- Automotive: ISO 26262–6:2018 §5 (Software development); ISO 21448:2022 §6; ISO 34503 (ODD attributes)
- Medical: FDA AI/ML SaMD Action Plan (2021); FDA PCCP Guidance (2023); IEC 62304:2006+A1:2015 §5.1
- EU: EU AI Act Article 10 (Data and data governance); EU AI Act Article 17 (Quality management)
- Government/Defense: 32 CFR Part 2002 (CUI); 22 CFR 120.17 (ITAR deemed exports); 15 CFR 762 (EAR recordkeeping)
- Cross-Domain: NIST AI RMF MAP–1 through MAP–5; NIST SP 800–53 Rev. 5

2.1.0 Operational Design Domain

[R.AI–1.0] OPERATIONAL DESIGN DOMAIN

The AI system shall declare an Operational Design Domain (ODD) that defines the operational conditions, inputs, users, subjects, regulatory context, and constraints under which the system is designed and validated to operate. The ODD declaration shall begin with an Operational Claim summary statement and shall address the eight attribute categories defined in Table 2.1.0–A.

Rationale: *AI systems trained against representative data perform reliably within the conditions represented in that data and become unreliable outside those conditions. Without an explicit declaration of operational scope, the certification claim is unbounded; the certificate appears to attest to system behavior under all conditions when verification evidence supports only a subset. ODD declaration establishes the operational envelope as a contractual element of the certification, scoping the certification claim to declared conditions and providing the basis from which downstream requirements are derived: dataset partitioning (AI–1.1), bias evaluation across subject populations (AI–2), operational input distribution testing (AI–3.3), output validity boundaries (AI–4.4), training distribution characterization (AI–6.1), and operational role verification (AI–9.6). ODD is the operational complement to system classification: classification establishes what the system is; ODD establishes where, for whom, and how it is intended to operate.*

Operational Claim: The ODD declaration shall begin with a single-sentence Operational Claim summarizing the scope of the certification: “*This AI system is designed and validated to [function] for [subject population] under [operational conditions], operated by [user role] within [regulatory regime].*” The Operational Claim is the elevator-summary of the ODD; the attribute categories below specify it in detail.

The ODD declaration shall address the eight attribute categories defined in Table 2.1.O-A. Exclusions shall be declared using the five exclusion sub-categories defined in Table 2.1.O-B. Threshold categories (Section 1.5) and operational input distributions (AI-3.3) shall be derived from and consistent with the declared ODD.

Table 2.1.O-A: Required ODD Attribute Categories

Category	Description	Examples	Downstream Requirements
Operational Environment	Physical, spatial, atmospheric, and temporal conditions of operation	Geographic scope, weather and atmospheric conditions, illumination and time-of-day ranges, terrain or surface types, ambient temperature ranges, care setting (medical), facility class	AI-3.3, AI-6.1
System Inputs	Sensor and data source characteristics, supported health states, input quality envelopes, and defined system behavior at degradation thresholds	Sensor types and configurations, sensor health states under which operation is supported, data source provenance, input quality bounds, defined behavior (degraded mode, disengage, fallback, non-diagnostic message) when degradation thresholds are reached	AI-3.3, AI-6.1, AI-9.4
User Population	Operator qualifications, training, currency, authorized roles, and role within operational workflow	Required operator certifications, system-specific training, currency requirements, role-based authorization levels, supported team configurations, user role in the decision workflow (autonomous, decision support, advisory)	AI-9.1, AI-9.3
Subject Population	Populations or entities the AI's decisions act upon, with	Validated patient demographics (medical), vehicle/pedestrian/cyclist populations the perception	AI-2, AI-3.3, AI-4.4

Category	Description	Examples	Downstream Requirements
	validated subject characteristics and excluded subject conditions	system handles (automotive), aircraft and obstacle classes (aviation), entity classes for surveillance or targeting systems	
Operational Phases	Intended use cases, operational phase transitions, and phase-level constraints	Supported workflow stages or mission segments, mode transition conditions, supported task categories, maximum phase or session duration, operational tempo (frequency, duty cycle, throughput per operator)	AI-9.4, AI-9.6
Performance Envelope	Per-decision performance limits and timing requirements	Detection ranges, sensitivity and specificity, accuracy thresholds, decision latency, throughput bounds, confidence calibration error, computational resource availability	AI-3.3, AI-4.2, AI-4.4, AI-8.3
Regulatory and Operational Authorization	Applicable regulatory regime, required certifications, waivers, authorization conditions, and ODD evolvability under change control	Governing regulatory framework (FAA Part 107, FDA SaMD class, ISO 26262 ASIL, EU AI Act risk class), required waivers or special authorizations, jurisdictional limitations, applicable industry standards, ODD evolvability declaration (locked at certification, or evolvable under defined change control mechanism such as FDA PCCP)	All of Section 2 (foundational scoping)
Exclusions	Explicit out-of-scope conditions under which the	See Table 2.1.0-B for required exclusion sub-categories	AI-4.4, AI-6.3, AI-9.1

Category	Description	Examples	Downstream Requirements
	system shall not engage		

Note on Operational Phases vs. Performance Envelope: Operational Phases captures phase-level or mission-level constraints (duration, sequence, transitions, tempo). Performance Envelope captures per-decision performance (latency, throughput, accuracy at the inference level). Time-bounded attributes that describe the workflow or mission belong in Operational Phases; time-bounded attributes that describe individual inferences belong in Performance Envelope.

Note on Subject Population: Not all systems have a distinct subject population from their user population. Where the AI acts on the user themselves (e.g., a personal recommendation system with no third-party subject), the Subject Population entry may state "Subject = User; see User Population." Where subjects are abstract objects rather than people (e.g., obstacle detection in drones), Subject Population still applies and should declare the validated object classes.

Table 2.1.O-B: Required Exclusion Sub-Categories

Sub-Category	Description	Example
Excluded Environments	Environmental conditions outside the operational envelope	Night operations, heavy precipitation, visibility below threshold, temperature extremes, terrain types not supported, deployment outside cleared jurisdiction
Excluded Use Cases	Mission types or operational scenarios outside intended use	Aerobatic maneuvers, formation operations, payload drop, multi-system coordination, mission durations beyond validated bounds, screening use of a diagnostic-only system
Excluded Operator Conditions	Operator states, qualifications, or staffing configurations not supported	Operators without current certification, operators below minimum currency, staffing below required minimum, non-radiologist users of a radiology decision support system
Excluded Subject Conditions	Subject populations or conditions outside validated scope	Pediatric patients in an adult-validated medical system, pedestrians in wheelchairs or carrying large objects in a perception system not validated for

Sub-Category	Description	Example
		those cases, subject conditions explicitly excluded from training data
Excluded System States	System or sensor degradation conditions that trigger automatic disengage or fallback	Sensor degradation beyond defined thresholds, communication loss conditions, computational resource exhaustion, image quality below threshold, GPS-denied conditions

[V.AI-1.0] OPERATIONAL DESIGN DOMAIN

The AI system shall verify ODD declaration by inspection.

The inspection shall confirm that the ODD declaration includes a single-sentence Operational Claim summary, that the declaration addresses all eight attribute categories required for the system's classification per Section 1.2.3 Table 1-4, that quantitative bounds are documented where the attribute is measurable, that exclusions are explicitly enumerated using the five sub-categories defined in Table 2.1.O-B, that the ODD declaration is approved at the authority level specified for the system's classification, and that downstream artifacts (operational input distributions per AI-3.3, training distribution characterization per AI-6.1, output validity boundaries per AI-4.4, bias evaluation populations per AI-2) are derived from and consistent with the declared ODD.

Success Criteria: The verification shall be considered successful when the inspection shows that the Operational Claim is present and consistent with the detailed ODD, ODD declaration addresses all required attribute categories, quantitative bounds are documented where measurable, exclusions are explicitly enumerated using the required sub-categories, declaration is approved at the required authority level, and downstream artifacts are traceable to the declared ODD.

ODD Maintenance: The ODD declaration shall be reviewed at the periodic intervals defined for operational role verification (AI-9.6). Identified divergence between declared ODD and actual operational use shall trigger formal ODD revision under the same authority structure as initial declaration. For systems declared as evolvable under a change control mechanism (e.g., FDA PCCP), ODD evolution shall follow the approved change control plan; ODD changes outside the approved plan shall require recertification.

2.1.1 Dataset Separation

[R.AI-1.1] DATASET SEPARATION

The AI system shall maintain separation between training, validation, and test datasets.

Rationale: *Data leakage where information from test or validation sets contaminates training data produces artificially inflated performance metrics. A model that has “seen” test data during training will perform well on that data but may fail on novel operational inputs. Strict separation with technical enforcement ensures performance estimates reflect true generalization capability.**

[V.AI-1.1] DATASET SEPARATION

The AI system shall verify dataset separation by inspection.

The inspection shall confirm dataset partitions are documented with unique identifiers, partition boundaries are enforced with technical controls (e.g., separate storage locations, access controls, automated pipeline checks), no data leakage exists between partitions, and partition integrity is verified prior to model training and evaluation.

Success Criteria: The verification shall be considered successful when the inspection shows that dataset partitions are documented with unique identifiers, partition boundaries are enforced with technical controls, no data leakage exists between partitions, and partition integrity is verified prior to training and evaluation.

2.1.2 Partition Use Restriction

[R.AI-1.2] PARTITION USE RESTRICTION

The AI system shall ensure each dataset partition is used only for its intended purpose.

Rationale: *Even with proper separation, using partitions incorrectly (e.g., using validation data for final testing, or repeatedly evaluating against test data during development) compromises the integrity of performance estimates. Restricting partition use to intended purposes and logging all access ensures data integrity is maintained throughout the development lifecycle.*

[V.AI-1.2] PARTITION USE RESTRICTION

The AI system shall verify partition use restriction by inspection and analysis.

The inspection shall confirm intended use for each partition is documented (training for model fitting, validation for hyperparameter tuning and model selection, test for final performance evaluation).

The analysis shall confirm access controls restrict partition use to authorized processes, audit logs capture dataset access and usage, and violation of intended use triggers alert and investigation.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that intended use for each partition is documented, access controls restrict partition use to authorized processes, audit logs capture access and usage, and violations trigger alerts and investigation.

2.1.3 Data Provenance

[R.AI-1.3] DATA PROVENANCE

The AI system shall document data provenance for all datasets used in model development.

Rationale: *AI model behavior is a direct function of training data. Without documented provenance, the source, quality, and potential biases in training data cannot be assessed. Provenance documentation enables root cause analysis when model failures occur, supports retraining decisions, and provides traceability required for safety-critical systems.*

[V.AI-1.3] DATA PROVENANCE

The AI system shall verify data provenance by inspection.

The inspection shall confirm data sources are identified and documented for each dataset, data collection methods are documented (including sensors, sampling rates, time periods, and environmental conditions), data transformations and preprocessing steps are recorded (including normalization, augmentation, filtering, and feature extraction), and provenance documentation is traceable to source records.

Success Criteria: The verification shall be considered successful when the inspection shows that data sources are identified and documented, collection methods are documented, transformations and preprocessing are recorded, and provenance is traceable to source records.

2.1.4 Ground Truth Labeling Quality

[R.AI-1.4] GROUND TRUTH LABELING QUALITY

The AI system shall validate ground truth labeling quality for all supervised learning datasets.

Rationale: Supervised learning models learn to predict labels if labels are incorrect, the model learns incorrect behavior. Label errors are particularly dangerous because they directly encode wrong answers into the model. For safety-critical applications, label quality must be validated through independent verification, as label errors could cause the model to make dangerous predictions with high confidence.

[V.AI-1.4] GROUND TRUTH LABELING QUALITY

The AI system shall verify ground truth labeling quality by inspection and analysis.

The inspection shall confirm labeling methodology and labeler qualifications are documented.

The analysis shall confirm inter-rater reliability metrics meet defined thresholds for multi-labeler datasets (e.g., Cohen's Kappa ≥ 0.8 for safety-critical applications), label accuracy is validated through spot-checking or independent verification for safety-critical applications, ambiguous or disputed labels are identified and resolved with documented rationale, and label error rate is estimated and its impact on model performance is assessed.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that labeling methodology and qualifications are documented, inter-rater reliability meets defined thresholds, label accuracy is validated for safety-critical applications, ambiguous labels are resolved with rationale, and label error rate impact is assessed.

2.1.5 PRE-INGESTION CLASSIFICATION SCREENING

[R.AI-1.5] PRE-INGESTION CLASSIFICATION SCREENING

The AI system shall screen all data sources for information classification markings prior to ingestion into model training, fine-tuning, or retrieval-augmented generation pipelines.

Rationale: AI training pipelines ingest data from diverse sources including government databases, technical repositories, contractor deliverables, sensor archives, and publicly available datasets. Any of these sources may contain data subject to information handling controls (CUI, ITAR, EAR, or other classification designations). Once controlled data is ingested into model training, the classification marking is permanently separated from the content. Pre-ingestion screening is the only reliable point at which to prevent classification loss, as no post-training mechanism exists to identify or remove controlled content from model parameters. This requirement establishes screening as a mandatory gate in

the data pipeline, analogous to the partition integrity verification required by AI-1.1 prior to every training run.

[V.AI-1.5] PRE-INGESTION CLASSIFICATION SCREENING

The AI system shall verify pre-ingestion classification screening by inspection and analysis.

The inspection shall confirm that a documented screening process exists for all data sources entering AI training, fine-tuning, or RAG pipelines; that screening criteria include CUI markings (per 32 CFR Part 2002 marking requirements), ITAR controlled technical data identifiers, EAR Export Control Classification Numbers (ECCNs), and any organization-specific classification markings; and that screening results are documented for each data source with disposition (approved, excluded, or approved with handling controls).

The analysis shall confirm that automated screening tools are validated against known-positive controlled content (detection rate $\geq 95\%$ for marked content); that manual review procedures are defined for data sources where automated screening is insufficient; and that screening coverage is complete, and no data enters the training pipeline without a documented screening disposition.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that a documented screening process exists, screening criteria cover all applicable classification regimes, screening results are documented with disposition for each data source, automated tools are validated against known-positive content, manual review procedures exist for edge cases, and no data enters the pipeline without screening disposition.

2.1.6 INFORMATION CLASSIFICATION INHERITANCE

[R.AI-1.6] INFORMATION CLASSIFICATION INHERITANCE

The AI system shall document the information classification level of all training data and shall apply the principle that AI model artifacts and outputs inherit the highest classification level of the training data unless the training pipeline demonstrates, with verifiable evidence, that controlled content was excluded.

Rationale: *In traditional information systems, data classification follows the data: a CUI document copied to a new system requires the new system to handle the copy as CUI. AI systems break this principle because the data transformation from documents to model weights destroys the classification metadata while retaining the controlled content. Without an explicit inheritance rule, the default assumption becomes that AI outputs are uncontrolled, which is incorrect when the training data included controlled content. This requirement establishes classification inheritance*

as the default, placing the burden of proof on the pipeline to demonstrate that controlled content was excluded rather than on the consumer to determine whether generated outputs contain controlled information. This is consistent with the derivative classification principle in information security: products derived from classified sources inherit the classification of the source material.

[V.AI-1.6] INFORMATION CLASSIFICATION INHERITANCE

The AI system shall verify information classification inheritance by inspection and analysis.

The inspection shall confirm that the classification level of all training datasets is documented in the data provenance record required by AI-1.3; that model artifacts (trained weights, serialized models, deployment packages) are marked with the inherited classification level; and that the classification inheritance determination is documented with justification, either inheriting the highest training data classification or certifying exclusion of controlled content with evidence.

The analysis shall confirm that the classification inheritance chain is complete from data sources through model artifacts to deployment endpoint; that any claim of controlled content exclusion is supported by the screening evidence required by AI-1.5; and that downstream systems receiving model outputs are authorized to handle information at the inherited classification level.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that training data classification is documented in provenance records, model artifacts carry inherited classification markings, inheritance determination is documented with justification, the inheritance chain is complete, exclusion claims are supported by screening evidence, and downstream systems are authorized for the inherited classification level.

2.1.7 OUTPUT CLASSIFICATION AWARENESS

[R.AI-1.7] OUTPUT CLASSIFICATION AWARENESS

The AI system shall implement mechanisms to ensure that consumers of AI-generated outputs are aware of the information classification handling requirements applicable to those outputs.

Rationale: *When an AI model trained on or exposed to controlled content generates outputs, those outputs may contain reconstructed controlled information. If the model operates at an inherited classification level per AI-1.6, all outputs must be handled at that level. However, the primary risk is that end users receive AI-generated content without any indication of applicable handling requirements. The user treats the output as uncontrolled general knowledge, copies*

it to uncontrolled systems, shares it with unauthorized recipients, or disseminates it to foreign persons, each constituting a potential violation of applicable regulations. This requirement ensures that classification awareness propagates to the point of consumption, not just the point of model storage. For ITAR technical data, failure to maintain this awareness constitutes a potential deemed export violation (22 CFR 120.17). For CUI, it violates dissemination controls established by the CUI Registry.

[V.AI-1.7] OUTPUT CLASSIFICATION AWARENESS

The AI system shall verify output classification awareness by test and analysis.

The test shall confirm that AI-generated outputs include or are accompanied by applicable classification handling indicators appropriate to the output channel (e.g., CUI markings on documents, banner markings on display interfaces, metadata tags on programmatic outputs); that output classification indicators are consistent with the inherited classification level determined under AI-1.6; that classification indicators cannot be trivially stripped or bypassed by end users during normal operational use; and that when AI outputs are consumed by downstream systems or pipelines, classification metadata is propagated programmatically.

The analysis shall confirm that output classification awareness mechanisms are appropriate for all operational output channels (display, document generation, API response, data export); that operator training materials address the classification handling requirements for AI-generated outputs; and that the output classification mechanism addresses the case where a model is queried by users at different authorization levels, ensuring outputs are appropriate to the requesting user's authorization.

Success Criteria: The verification shall be considered successful when the test and analysis show that outputs include classification handling indicators, indicators are consistent with the inherited classification level, indicators are not trivially bypassed, downstream propagation is implemented, all output channels are covered, operator training addresses output handling, and user authorization levels are respected in output generation.

2.1.8 CLASSIFICATION COMPLIANCE AUDIT TRAIL

[R.AI-1.8] CLASSIFICATION COMPLIANCE AUDIT TRAIL

The AI system shall maintain an audit trail documenting classification screening decisions, inheritance determinations, output classification events, and any identified or suspected classification incidents throughout the AI system lifecycle.

Rationale: Regulatory compliance for CUI, ITAR, and EAR requires demonstrable evidence of proper handling. Traditional information systems maintain access logs

and handling records. AI systems require analogous records covering the AI-specific aspects of information classification: what data was screened and what was the disposition, what classification level was inherited and on what basis, what outputs were generated at what classification level, and whether any incidents of suspected classification loss or improper dissemination occurred. Without this audit trail, an organization cannot demonstrate compliance during regulatory review, cannot scope the impact of a discovered classification incident, and cannot support the continuous improvement of classification controls. This requirement is the classification-specific complement to the human-AI interaction logging required by AI-9.5 and the OOD event logging required by AI-6.4.

[V.AI-1.8] CLASSIFICATION COMPLIANCE AUDIT TRAIL

The AI system shall verify classification compliance audit trail by inspection.

The inspection shall confirm that pre-ingestion screening decisions are logged with timestamps, data source identifiers, screening method, and disposition; that classification inheritance determinations are logged with justification and authorizing authority; that output classification events are logged at a level sufficient to support incident investigation (including query context, output classification level applied, and output channel); that suspected classification incidents are logged with timestamps, discovery method, affected data scope estimate, and remediation actions; and that audit trail records are retained for a period consistent with applicable regulatory requirements (minimum retention periods per 32 CFR Part 2002 for CUI, ITAR record retention requirements per 22 CFR 122.5, and EAR record retention requirements per 15 CFR 762).

Success Criteria: The verification shall be considered successful when the inspection shows that screening decisions are logged, inheritance determinations are logged with justification, output classification events are logged, suspected incidents are logged with remediation actions, and retention periods meet applicable regulatory requirements.

2.2 AI-2: ADDRESSING AI BIAS

[R.AI-2] ADDRESSING AI BIAS

The AI system will implement bias detection, prevention, and mitigation mechanisms to provide equitable and unbiased decision-making across all applicable operational scenarios and all applicable user populations.

AI systems trained on historical data can encode and amplify biases present in that data, leading to systematic performance disparities across operational conditions or user populations. Bias in this context refers to any systematic deviation in AI

performance whether across environmental conditions, sensor degradation states, or user characteristics. This requirement addresses the gap between traditional software verification (which assumes uniform behavior) and AI systems whose behavior may vary systematically based on input characteristics. Bias evaluation populations shall be derived from the Subject Population declared under AI-1.O.

Applicable Domain Standards

- General: ISO/IEC TR 24027:2021; ISO/IEC 22989:2022 §3.5.4
- Aviation: FAA AI Safety Roadmap (equitable operation across crew demographics) [verify section]
- Medical: FDA Draft Guidance on AI/ML Device Bias (2024) [verify]; FDA Digital Health Policy Navigator
- EU: EU AI Act Article 10(2)(f) (bias detection and correction); EU AI Act Article 15(3) (accuracy across groups)
- Consumer/Employment: EEOC AI Technical Assistance (2023) (where employment-adjacent)
- Cross-Domain: NIST AI RMF MANAGE-2.3; NIST SP 1270 (Taxonomy of AI Bias, 2022)

2.2.1 Bias-Free Baseline Performance

[R.AI-2.1] BIAS-FREE BASELINE PERFORMANCE

The AI system shall be baselined with bias-free performance across user classes and operational contexts.

Rationale: Establishing baseline performance prior to deployment provides a reference point for continuous monitoring and enables detection of bias drift during operations. Without documented baselines, degradation in performance cannot be objectively measured or addressed.

[V.AI-2.1] BIAS-FREE BASELINE PERFORMANCE

The AI system shall verify bias-free baseline performance by analysis and test.

The analysis shall confirm bias-free performance criteria are defined for all applicable user attributes and operational contexts relevant to the AI function.

The test shall provide results congruent with defined bias-free performance criteria using validation datasets representative of operational conditions.

Success Criteria: The verification shall be considered successful when the analysis and test show that bias-free baseline criteria are defined, test results meet defined criteria prior to operational deployment, and a validation report captures baseline values, threshold criteria, and threshold justification.

Applicable Domain Standards

- General: ISO/IEC TR 24027:2021 §6 (Bias assessment approaches); ISO/IEC 22989:2022 §3.5.4 (Bias definition)
- Aviation: FAA AI Safety Roadmap (equitable performance across operational demographics) [verify section]
- Automotive: UNECE WP.29 R157 (ALKS) performance consistency requirements [verify paragraph]
- Medical: FDA Draft Guidance on AI/ML Device Bias (2024) [verify]; FDA Digital Health Policy Navigator
- EU: EU AI Act Article 10(2)(f) (bias examination and correction); EU AI Act Article 10(3) (training data representativeness); EU AI Act Article 15(3) (performance across demographic groups)
- Consumer/Employment (where applicable): EEOC AI Technical Assistance (2023)
- Cross-Domain: NIST AI RMF MANAGE-2.3; NIST SP 1270:2022 (Taxonomy of AI Bias)

2.2.2 Bias-Free Training Data

[R.AI-2.2] BIAS-FREE TRAINING DATA

The AI system shall use training datasets that account for underrepresented groups, historical biases, and labeling errors.

Rationale: *Training data quality directly determines AI system behavior. Datasets with underrepresented groups, historical biases, or labeling errors will produce models that perpetuate those biases.*

[V.AI-2.2] BIAS-FREE TRAINING DATA

The AI system shall verify bias-free training data by inspection.

The inspection shall confirm training datasets document representation across applicable user classes and operational contexts, known historical biases are identified and addressed, and labeling methodology includes quality controls.

Success Criteria: The verification shall be considered successful when the inspection shows that training data representation is documented, historical biases are identified and addressed, and labeling quality controls are documented.

2.2.3 Training Dataset Validation

[R.AI-2.3] TRAINING DATASET VALIDATION

The AI system shall validate training datasets that could impact mission-critical decisions.

Rationale: *Validation of training datasets prior to model development confirms that bias-free training data requirements have been met and provides documented evidence for verification closure.*

[V.AI-2.3] TRAINING DATASET VALIDATION

The AI system shall verify training dataset validation by inspection and analysis.

The inspection shall confirm training datasets do not contain data that contributes to biases such as underrepresented groups, historical biases, or labeling errors.

The analysis shall quantify representation across applicable user classes and operational contexts and confirm absence of known bias patterns. Where absence cannot be confirmed, the analysis shall include a risk assessment due to included bias and provide mitigation strategy for reducing the risk.

Success Criteria: The verification shall be considered successful when the inspection confirms training datasets do not contain bias-contributing data, and the analysis confirms either absence of known bias patterns or documented risk assessment with mitigation strategy.

2.2.4 Continuous Bias Monitoring

[R.AI-2.4] CONTINUOUS BIAS MONITORING

The AI system shall implement continuous bias monitoring appropriate to the operational context.

Rationale: *AI system bias can emerge or amplify during operations due to data drift, changing operational contexts, or feedback loops. Continuous monitoring enables early detection of bias before it impacts safety or operational success. Monitoring frequency and methods should scale with system classification and decision criticality.*

[V.AI-2.4] CONTINUOUS BIAS MONITORING

The AI system shall verify continuous bias monitoring by test.

The test shall confirm continuous bias monitoring is operational and metrics are maintained within approved thresholds.

Success Criteria: The verification shall be considered successful when the test shows that continuous bias monitoring is operational, and metrics are maintained within approved thresholds.

2.2.5 Bias Impact Assessment

[R.AI-2.5] BIAS IMPACT ASSESSMENT

The AI system shall conduct bias impact assessments for all AI decisions affecting human safety, resource allocation, mission planning, and operational procedures.

Rationale: *Not all AI decisions carry equal risk of harm from bias. Impact assessments ensure that bias mitigation efforts are prioritized for decisions with the greatest potential consequences for safety and operational success.*

[V.AI-2.5] BIAS IMPACT ASSESSMENT

The AI system shall verify bias impact assessments by analysis and test.

The analysis shall document bias impact assessments for each decision category and ensure assessment methodology is traceable to defined criteria.

The test shall confirm assessment results inform mitigation strategies prior to deployment.

Success Criteria: The verification shall be considered successful when the analysis and test show that bias impact assessments are documented for each decision category, assessment methodology is traceable to defined criteria, and results inform mitigation strategies prior to deployment.

2.2.6 Bias Indicator Definition

[R.AI-2.6] BIAS INDICATOR DEFINITION

The AI system shall define bias indicators and thresholds for each AI function.

Rationale: *Bias detection requires defined metrics (indicators) and acceptance boundaries (thresholds). Without explicit definition, bias detection is subjective and unverifiable.*

[V.AI-2.6] BIAS INDICATOR DEFINITION

The AI system shall verify bias indicator definition by inspection.

The inspection shall confirm bias indicators are defined for each AI function, threshold values are defined and justified for each indicator, and threshold derivation rationale is documented.

Success Criteria: The verification shall be considered successful when the inspection shows bias indicators are defined, thresholds are defined and justified, and derivation rationale is documented.

2.2.7 Bias Threshold Alerting

[R.AI-2.7] BIAS THRESHOLD ALERTING

The AI system shall generate alerts when bias indicators exceed predefined thresholds.

Rationale: *Detection without notification provides no opportunity for response. Alerts enable operators and systems to respond to detected bias conditions.*

[V.AI-2.7] BIAS THRESHOLD ALERTING

The AI system shall verify bias threshold alerting by test.

The test shall confirm alerts are generated when bias indicators exceed thresholds and alerts are delivered to designated recipients within specified latency limits.

Success Criteria: The verification shall be considered successful when the test shows alerts are generated when thresholds are exceeded and alerts are delivered within latency limits.

2.2.8 Bias Threshold Response

[R.AI-2.8] BIAS THRESHOLD RESPONSE

The AI system shall provide bias-corrected decision pathways and alternative recommendations when bias indicators exceed predefined thresholds.

Rationale: *Detection and alerting without corrective action provides no protection. This requirement ensures the system has predefined corrective actions when bias is detected.*

[V.AI-2.8] BIAS THRESHOLD RESPONSE

The AI system shall verify bias threshold response by test.

The test shall confirm bias-corrected pathways are invoked when thresholds are exceeded and alternative recommendations are presented.

Success Criteria: The verification shall be considered successful when the test shows bias-corrected pathways are invoked and alternative recommendations are presented when thresholds are exceeded.

2.2.9 Bias Event Logging

[R.AI-2.9] BIAS EVENT LOGGING

The AI system shall include in logs bias detection events, corrective actions taken, and decision rationale for accountability and post-mission analysis.

Rationale: *Logs provide accountability and enable post-mission analysis to improve future AI systems. Without comprehensive logging, bias events cannot be reconstructed for root cause analysis or used to improve subsequent model versions.*

[V.AI-2.9] BIAS EVENT LOGGING

The AI system shall verify bias event logging by inspection.

The inspection shall confirm logs record bias detection events with timestamps, logs capture metrics, decision context, mitigation actions, and operator inputs, logs are immutable and retrievable for post-mission analysis, and log schema documentation is complete.

Success Criteria: The verification shall be considered successful when the inspection shows that logs record bias detection events with timestamps, logs capture all required information, logs are immutable and retrievable, and log schema documentation is complete.

2.2.10 Safety-Critical Bias Prohibition

[R.AI-2.10] SAFETY-CRITICAL BIAS PROHIBITION

The AI system shall not allow bias to adversely impact safety-critical operations or human safety.

Rationale: *Bias affecting safety-critical functions poses unacceptable risk. This requirement establishes the fundamental constraint that detected bias must be prevented from affecting safety-critical decisions, with escalation and override as the enforcement mechanism.*

[V.AI-2.10] SAFETY-CRITICAL BIAS PROHIBITION

The AI system shall verify safety-critical bias prohibition by analysis and test.

The analysis shall identify all AI decision points that affect safety-critical operations.

The test shall confirm that detected bias at safety-critical decision points triggers protective action preventing biased outputs from affecting safety-critical operations.

Success Criteria: The verification shall be considered successful when the analysis identifies safety-critical decision points and the test confirms bias detection triggers protective action at those points.

2.2.11 Safety-Critical Bias Escalation

[R.AI-2.11] SAFETY-CRITICAL BIAS ESCALATION

The AI system shall escalate any bias detection that would adversely impact safety-critical operations or human safety for immediate human review and potential system override with documented rationale.

Rationale: *Bias affecting safety-critical functions requires immediate human intervention. Autonomous continuation in the presence of detected safety-critical bias could result in harm. This requirement ensures human authority is maintained over safety-critical AI decisions per AI-9.*

[V.AI-2.11] SAFETY-CRITICAL BIAS ESCALATION

The AI system shall verify safety-critical bias escalation by test.

The test shall simulate safety-critical bias scenarios and confirm mandatory human review states are triggered, autonomous execution is blocked until review is complete, system override capability is functional, and escalation logs capture review decisions and rationale.

Success Criteria: The verification shall be considered successful when the test shows that safety-critical bias scenarios trigger mandatory human review, autonomous execution is blocked until review is complete, system override capability is functional and documented, and escalation logs capture review decisions and rationale.

2.2.12 Bias Edge Case Definition

[R.AI-2.12] BIAS EDGE CASE DEFINITION

The AI system shall define edge-case datasets and stress scenarios for bias validation.

Rationale: *Edge cases and stress conditions represent operational boundaries where bias is most likely to emerge but least likely to be represented in training*

data. Explicit definition of edge cases ensures systematic coverage during validation rather than ad-hoc testing.

[V.AI-2.12] BIAS EDGE CASE DEFINITION

The AI system shall verify bias edge case definition by inspection.

The inspection shall confirm edge-case datasets are defined and documented, stress scenarios are defined and documented, and edge cases address operational boundaries and conditions underrepresented in training data.

Success Criteria: The verification shall be considered successful when the inspection shows that edge-case datasets and stress scenarios are defined and documented.

2.2.13 Bias Edge Case Validation

[R.AI-2.13] BIAS EDGE CASE VALIDATION

The AI system shall validate bias performance under defined edge cases and stress scenarios.

Rationale: *Bias may not manifest under nominal conditions but emerge at operational boundaries or under stress. Validation across edge cases ensures the system maintains bias-free performance even in conditions that may not be well-represented in training data.*

[V.AI-2.13] BIAS EDGE CASE VALIDATION

The AI system shall verify bias edge case validation by test.

The test shall confirm bias metrics remain within acceptable bounds under defined edge cases and stress scenarios, or threshold exceedances trigger defined mitigation.

Success Criteria: The verification shall be considered successful when the test shows bias metrics remain within acceptable bounds under edge cases and stress scenarios, or exceedances trigger defined mitigation.

2.3 AI-3: ML TEST COVERAGE

[R.AI-3] ML TEST COVERAGE

The AI system will demonstrate adequate test coverage for machine learning models using methods appropriate to data-driven systems where traditional code coverage metrics do not apply.

Traditional software testing relies on code coverage metrics (statement coverage, branch coverage, MC/DC) to demonstrate that the software has been adequately exercised. These metrics are meaningless for ML models there is no “code” to cover in a neural network’s weights. AI systems require alternative coverage concepts based on input space coverage, operational scenario coverage, and behavioral testing. This requirement addresses the “no test coverage equivalent” gap identified in Section 1.2.4.2.

Applicable Domain Standards

- General: ISO/IEC/IEEE 29119-11:2020 (Testing of AI-based systems); ISO/IEC 5338:2023 §6.4
- Aviation: DO-178C §6 (Verification Process); EUROCAE ED-324/SAE AS6983 (AI/ML aviation) [verify]
- Automotive: ISO 26262-6:2018 §9 (Software unit verification); ISO 21448:2022 §7 (Evaluation)
- Medical: IEC 62304:2006+A1:2015 §5.7 (Software system testing)
- EU: EU AI Act Article 15 (Accuracy, robustness and cybersecurity)
- Cross-Domain: NIST AI RMF MEASURE-2; IEEE 7010-2020

2.3.1 Test Coverage Metric Definition

[R.AI-3.1] TEST COVERAGE METRIC DEFINITION

The AI system shall define test coverage metrics appropriate to each ML model type.

Rationale: *Different ML model types require different coverage approaches. For image classifiers, coverage might focus on variations in lighting, angle, and occlusion. For time-series predictors, coverage might address temporal patterns, seasonality, and anomalies. Metrics must be justified based on model architecture and mission criticality to ensure testing is meaningful rather than merely comprehensive in volume.*

[V.AI-3.1] TEST COVERAGE METRIC DEFINITION

The AI system shall verify test coverage metric definition by inspection.

The inspection shall confirm test coverage metrics are defined for each ML model, metrics are justified based on model type and mission criticality, coverage metrics address input space, edge cases, and operational scenarios, and rationale for metric selection is documented.

Success Criteria: The verification shall be considered successful when the inspection shows that test coverage metrics are defined for each ML model, metrics are justified based on model type and criticality, coverage addresses input space, edge cases, and operational scenarios, and rationale is documented.

2.3.2 Test Coverage Threshold Achievement

[R.AI-3.2] TEST COVERAGE THRESHOLD ACHIEVEMENT

The AI system shall achieve defined test coverage thresholds prior to operational deployment.

Rationale: *Defining metrics without achieving them provides no assurance. Coverage thresholds should be established based on system classification Safety-Critical AI requires higher coverage thresholds than Operational Support AI. Coverage gaps must be identified and dispositioned with documented rationale, either by additional testing or by risk acceptance with mitigation.*

[V.AI-3.2] TEST COVERAGE THRESHOLD ACHIEVEMENT

The AI system shall verify test coverage threshold achievement by analysis and test.

The analysis shall confirm coverage thresholds are defined and documented with justification based on system classification and mission risk tolerance.

The test shall execute test cases to achieve required coverage levels, identify and document any coverage gaps, and disposition gaps through additional testing or documented risk acceptance.

Success Criteria: The verification shall be considered successful when the analysis and test show that coverage thresholds are defined and documented, test execution achieves required coverage levels, coverage gaps are identified and dispositioned, and coverage reports are retained as verification evidence.

2.3.3 Operational Input Distribution Testing

[R.AI-3.3] OPERATIONAL INPUT DISTRIBUTION TESTING

The AI system shall test ML models across representative operational input distributions.

Rationale: *ML models perform well on data similar to their training data but may fail on data that differs from training distributions. Testing must cover the full range of inputs expected during actual operations, including nominal conditions, off-nominal conditions, and mission phase transitions. Gaps between test and operational*

distributions represent untested regions where model behavior is uncertain. Operational input distributions shall be derived from the declared ODD (AI-1.0).

[V.AI-3.3] OPERATIONAL INPUT DISTRIBUTION TESTING

The AI system shall verify operational input distribution testing by test.

The test shall confirm operational input distributions are characterized and documented based on mission analysis and operational scenarios, test datasets represent the full range of expected operational inputs, model performance is validated across the operational input space, and gaps between test and operational distributions are identified and mitigated through additional test data collection, synthetic data generation, or documented risk acceptance.

Success Criteria: The verification shall be considered successful when the test shows that operational input distributions are characterized, test datasets represent the full operational range, model performance is validated across the input space, and distribution gaps are identified and mitigated.

2.3.4 Boundary and Edge Case Testing

[R.AI-3.4] BOUNDARY AND EDGE CASE TESTING

The AI system shall perform boundary and edge case testing for all ML models.

Rationale: *ML models often fail at input boundaries values at the edges of the training distribution or combinations of inputs not well-represented in training data. Edge cases are particularly important for safety-critical functions because they represent conditions where model behavior is least certain. Systematic boundary testing ensures the model behaves acceptably at operational limits.*

[V.AI-3.4] BOUNDARY AND EDGE CASE TESTING

The AI system shall verify boundary and edge case testing by test.

The test shall confirm input boundaries are defined for each ML model based on operational constraints and physical limits, edge cases are identified and documented through systematic analysis (e.g., equivalence partitioning, boundary value analysis adapted for ML), model behavior at boundaries is tested and acceptable, and edge case test results are recorded with pass/fail criteria.

Success Criteria: The verification shall be considered successful when the test shows that input boundaries are defined for each ML model, edge cases are identified and documented, model behavior at boundaries is tested and acceptable, and test results are recorded with pass/fail criteria.

2.4 AI-4: CONTINUOUS VALIDATION

[R.AI-4] CONTINUOUS VALIDATION

The AI system shall implement continuous validation processes to ensure model performance remains within acceptable bounds throughout operational deployment.

AI model performance can degrade during operations due to data drift, concept drift, or environmental changes. Continuous validation detects performance degradation before it impacts safety or operational success, enabling timely corrective action such as model retraining, threshold adjustment, or fallback to manual operations.

Applicable Domain Standards

- General: ISO/IEC 5338:2023 §6.4.14; ISO/IEC 22989:2022 §5.11.9
- Aviation: FAA AI Safety Roadmap (operational monitoring)
- Automotive: ISO 21448:2022 §13 (Field monitoring); UNECE WP.29 R157 (ALKS) §5.2 [verify]
- Medical: FDA PCCP Guidance (2023); FDA AI/ML-Based SaMD Action Plan §3 (Real-World Performance)
- EU: EU AI Act Article 17 (Quality management system); EU AI Act Article 72 (Post-market monitoring)
- Cross-Domain: NIST AI RMF MEASURE-3; NIST AI RMF MEASURE-4

2.4.1 Data Drift Monitoring

[R.AI-4.1] DATA DRIFT MONITORING

The AI system shall monitor for data drift that could cause inaccurate decisions or predictions.

Rationale: Data drift occurs when the statistical characteristics of production data diverge from the training data distribution. Sensor degradation, environmental changes, or novel operational scenarios can cause inputs to shift from what the model was trained on, leading to degraded prediction accuracy. Without drift detection, performance degradation may go unnoticed until a critical failure occurs.

[V.AI-4.1] DATA DRIFT MONITORING

The AI system shall verify data drift monitoring by test.

The test shall confirm data drift detection mechanisms are implemented and operational, drift is measured against baseline training data distributions using defined statistical methods (e.g., KL divergence, population stability index), drift metrics are logged with timestamps, and alerts are generated when drift exceeds defined thresholds.

Success Criteria: The verification shall be considered successful when the test shows that data drift detection mechanisms are operational, drift is measured against baseline distributions, drift metrics are logged with timestamps, and alerts are generated when drift exceeds defined thresholds.

2.4.2 Performance Threshold Maintenance

[R.AI-4.2] PERFORMANCE THRESHOLD MAINTENANCE

The AI system shall maintain model performance above defined thresholds throughout operation.

Rationale: *AI model performance can degrade over time even without data drift through concept drift, model staleness, or accumulated edge cases. Continuous performance monitoring against defined thresholds provides early warning of degradation before it impacts operational success or safety. Threshold values should be derived from mission risk tolerance per Section 4.*

[V.AI-4.2] PERFORMANCE THRESHOLD MAINTENANCE

The AI system shall verify performance threshold maintenance by analysis and test.

The analysis shall confirm performance thresholds are defined and justified based on mission risk tolerance, and that performance metrics are appropriate to the AI function (e.g., accuracy, precision, recall, F1 score, domain-specific metrics).

The test shall confirm performance metrics are continuously computed during operation, performance degradation below threshold triggers defined mitigation response, and performance logs are maintained for trend analysis.

Success Criteria: The verification shall be considered successful when the analysis and test show that performance thresholds are defined and justified, performance metrics are continuously computed, degradation below threshold triggers mitigation response, and performance logs support trend analysis.

2.4.3 Model Maintenance Criteria

[R.AI-4.3] MODEL MAINTENANCE CRITERIA

The AI system shall define criteria for model maintenance, retraining, or version updates when performance deviates beyond acceptable bounds.

Rationale: *When continuous monitoring detects performance degradation, the system must have predefined response procedures. Without documented criteria for when and how to update models, operators may allow degraded performance to continue or may apply ad-hoc fixes that introduce new risks. This requirement ensures maintenance decisions are systematic and traceable. Model maintenance changes also invoke the model integrity controls established under AI-7.4 and the deployment format validation established in Section 3.3; these three control points apply in concert whenever a deployed model is updated.*

[V.AI-4.3] MODEL MAINTENANCE CRITERIA

The AI system shall verify model maintenance criteria by inspection.

The inspection shall confirm retraining and update trigger criteria are documented, version control procedures exist for model updates, regression testing requirements are defined for updated models, and change history is maintained with rationale for each update.

Success Criteria: The verification shall be considered successful when the inspection shows that retraining/update trigger criteria are documented, version control procedures exist, regression testing requirements are defined, and change history with rationale is maintained.

2.4.4 Output Validity Boundaries

[R.AI-4.4] OUTPUT VALIDITY BOUNDARIES

The AI system shall define and enforce boundaries for output data validity.

Rationale: *AI systems can produce outputs that are technically valid (within data type constraints) but operationally invalid (physically impossible, outside safe operating ranges, or inconsistent with known constraints). Output boundary enforcement provides a safety envelope that catches erroneous outputs regardless of the internal cause whether from drift, adversarial inputs, or model failure. Operational limits that bound acceptable outputs shall be derived from the Performance Envelope and Exclusions declared under AI-1.0.*

[V.AI-4.4] OUTPUT VALIDITY BOUNDARIES

The AI system shall verify output validity boundaries by test.

The test shall confirm output boundaries are defined for each AI function based on physical constraints, operational limits, and safety margins. The test shall inject

inputs designed to produce boundary-case outputs and confirm outputs exceeding boundaries are flagged or rejected, boundary violations are logged with context, and out-of-bounds outputs do not propagate to downstream systems without human review or invocation of defined fallback behavior.

Success Criteria: The verification shall be considered successful when the test shows that output boundaries are defined for each AI function, outputs exceeding boundaries are flagged or rejected, boundary violations are logged, and out-of-bounds outputs do not propagate without human review or fallback behavior.

2.4.5 Periodic Model Validation

[R.AI-4.5] PERIODIC MODEL VALIDATION

The AI system shall perform model validation at defined intervals and maintain baseline validation scores.

***Rationale:** Continuous monitoring of individual metrics may not capture gradual, systemic degradation. Periodic comprehensive validation against baseline scores provides a systemic check that the model continues to perform as originally validated. Validation frequency should scale with system classification and operational tempo.*

[V.AI-4.5] PERIODIC MODEL VALIDATION

The AI system shall verify periodic model validation by test.

The test shall confirm validation frequency is defined and justified based on system classification and mission profile, validation executes automatically at specified intervals, validation scores are compared against baseline thresholds, and validation failures trigger defined response procedures.

Success Criteria: The verification shall be considered successful when the test shows that validation frequency is defined and justified, validation executes automatically at specified intervals, validation scores are compared against baselines, and validation failures trigger defined response procedures.

2.4.6 Operational Validation Without Ground Truth

[R.AI-4.6] OPERATIONAL VALIDATION WITHOUT GROUND TRUTH

The AI system shall define and justify proxy validation mechanisms for operational environments where ground truth data is unavailable or inaccessible.

Rationale: AI-4 requirements assume continuous validation against known correct outputs. This assumption holds during development and in connected operational

environments where labeled data or authoritative reference sources remain accessible. However, many safety critical operational contexts lack ground truth entirely: disconnected systems, autonomous platforms, deep space missions, submarine operations, or any environment where the “correct answer” cannot be independently determined in real time. In these environments, validation must rely on proxy mechanisms that provide evidence of continued reliable performance without direct comparison to ground truth. These proxies include output boundary enforcement (per AI-4.4), statistical drift detection against baseline distributions (per AI-4.1), independent cross validation using dissimilar methods (per AI-5.5), and temporal consistency checks against the system’s own recent output history. The selection and sufficiency of proxy mechanisms is context dependent. A system operating autonomously for hours faces different validation challenges than one operating autonomously for months. Projects shall document which proxy mechanisms apply, justify why those proxies are sufficient for the specific operational context and duration, and define the conditions under which proxy validation is no longer adequate and operations should be suspended or degraded.

[V.AI-4.6] OPERATIONAL VALIDATION WITHOUT GROUND TRUTH

The AI system shall verify operational validation without ground truth by inspection and analysis.

The inspection shall confirm the operational environment is characterized with respect to ground truth availability, proxy validation mechanisms are identified and documented for each AI function operating without ground truth, and the justification for proxy sufficiency is documented with traceability to operational context, mission duration, and system classification.

The analysis shall confirm selected proxy mechanisms collectively address performance degradation detection, output validity assurance, and behavioral consistency monitoring; that the analysis identifies conditions under which proxy validation is insufficient and documents the required system response (degraded mode, human escalation, or operational suspension); and that proxy validation coverage is assessed against the operational timeline to confirm adequacy for the expected duration of ground truth unavailability.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that the operational environment is characterized for ground truth availability, proxy mechanisms are documented and justified, proxy coverage addresses degradation detection, output validity, and behavioral consistency, insufficiency conditions are defined with required system response, and proxy adequacy is confirmed for the expected operational duration.

2.5 AI-5: HALLUCINATION PREVENTION

[R.AI-5] HALLUCINATION PREVENTION

The AI system shall implement mechanisms to detect, prevent, and mitigate AI outputs that are fabricated, unsupported by input data, or inconsistent with operational reality.

Rationale: *AI systems, particularly neural networks and generative models, can produce outputs that appear valid but are completely fabricated or contradicted by physical reality. NIST AI 600-1 defines hallucination as “the production of confidently stated but erroneous or false content.”** Unlike traditional software bugs that produce obviously wrong outputs (e.g., null values, exceptions), hallucinations appear plausible and may not trigger conventional error handling. This requirement addresses the *****“unpredictable outputs”***** gap identified in Section 1.2.4.2.*

Applicable Domain Standards

- General: NIST AI 600-1 (Generative AI Profile, 2024) §2.4; ISO/IEC 22989:2022 §5.11
- Automotive: ISO 21448:2022 §6 (Performance limitations)
- Medical: FDA Clinical Decision Support Software Guidance (2022) §IV (Evidence basis)
- EU: EU AI Act Article 15 (Robustness)
- Cross-Domain: NIST AI RMF MEASURE-2.7

2.5.1 Hallucination Criteria Definition

[R.AI-5.1] HALLUCINATION CRITERIA DEFINITION

The AI system shall define what constitutes a hallucination within the operational context.

Rationale: *Hallucination is context-dependent an output that would be a hallucination in one domain may be valid in another. For example, a predicted temperature of -270°C would be valid near absolute zero but a hallucination for cabin temperature. Defining mission-specific hallucination criteria enables automated detection and ensures operators understand when an output should be considered unreliable. Ground truth baselines provide authoritative reference points against which outputs can be validated.*

[V.AI-5.1] HALLUCINATION CRITERIA DEFINITION

The AI system shall verify hallucination criteria definition by inspection.

The inspection shall confirm hallucination criteria are defined for each AI function, criteria include output outside expected bounds, contradictory outputs, and outputs unsupported by inputs, definition is traceable to mission-specific operational constraints, examples of hallucination versus valid edge-case outputs are documented, and ground truth baselines are established for each AI function including authoritative data sources, physical constraints and conservation laws, cross-validation with independent sensors or systems, and operator-verified reference datasets.

Success Criteria: The verification shall be considered successful when the inspection shows that hallucination criteria are defined for each AI function, criteria address bounds violations, contradictions, and unsupported outputs, definitions are traceable to operational constraints, examples distinguish hallucinations from valid edge cases, and ground truth baselines are established.

2.5.2 Hallucination Detection

[R.AI-5.2] HALLUCINATION DETECTION

The AI system shall implement detection mechanisms for hallucinated outputs.

Rationale: *Once hallucination criteria are defined, automated detection mechanisms can identify suspect outputs before they propagate to downstream systems or influence decisions. Detection should occur at inference time, not just during post-mission analysis, to prevent real-time operational impact. Detection mechanisms may include consistency checks, physical constraint validation, cross-sensor comparison, or confidence thresholding.*

[V.AI-5.2] HALLUCINATION DETECTION

The AI system shall verify hallucination detection by test.

The test shall confirm detection algorithms are implemented for each AI function, detection mechanisms identify outputs meeting hallucination criteria, detection is performed prior to output delivery to downstream systems, and detection performance is validated against known hallucination test cases including injected hallucinations that violate physical constraints, contradict sensor inputs, or fall outside valid operating ranges.

Success Criteria: The verification shall be considered successful when the test shows that detection algorithms are implemented for each AI function, detection identifies outputs meeting hallucination criteria, detection occurs prior to

downstream delivery, and detection performance is validated against known test cases.

2.5.3 Hallucination Response

[R.AI-5.3] HALLUCINATION RESPONSE

The AI system shall prevent hallucinated outputs from being acted upon without human review.

Rationale: *Detection without response provides no protection. When a potential hallucination is detected, the system must prevent autonomous action based on that output. This ensures human judgment is applied before potentially erroneous AI outputs influence safety-critical decisions. The response should be proportionate to the criticality of the AI function and the confidence of hallucination detection.*

[V.AI-5.3] HALLUCINATION RESPONSE

The AI system shall verify hallucination response by test.

The test shall inject hallucinated outputs and confirm detected hallucinations are flagged and quarantined, autonomous execution is blocked for flagged outputs, human review workflow is triggered for flagged outputs, and override requires documented rationale.

Success Criteria: The verification shall be considered successful when the test shows that detected hallucinations are flagged and quarantined, autonomous execution is blocked, human review workflow is triggered, and override requires documented rationale.

2.5.4 Hallucination Logging

[R.AI-5.4] HALLUCINATION LOGGING

The AI system shall log all detected hallucinations for post-mission analysis.

Rationale: *Comprehensive logging of hallucination events enables post-mission analysis to identify patterns, root causes, and potential model improvements. Logs support trend analysis to determine if hallucination frequency is increasing (indicating model degradation) and provide evidence for retraining decisions. Logging also supports accountability and incident investigation.*

[V.AI-5.4] HALLUCINATION LOGGING

The AI system shall verify hallucination logging by inspection.

The inspection shall confirm hallucination events are logged with timestamps, inputs, and outputs, logs capture detection method and confidence, logs capture human review decision and rationale, and logs are retrievable for trend analysis and model improvement.

Success Criteria: The verification shall be considered successful when the inspection shows that hallucination events are logged with timestamps, inputs, and outputs, logs capture detection method, confidence, and human review decisions, and logs are retrievable for trend analysis.

2.5.5 Independent Output Validation

[R.AI-5.5] INDEPENDENT OUTPUT VALIDATION

The AI system shall validate AI outputs affecting safety-critical or mission-critical functions against an independent source prior to execution.

Rationale: *For safety-critical functions, relying solely on the AI system's internal consistency checks is insufficient as an AI can produce confidently wrong outputs that pass its own validation. Independent validation using dissimilar methods, sensors, or processing provides a defense-in-depth layer that can catch errors the primary AI cannot detect. The independent source must not share common failure modes with the primary AI to ensure that a single fault (e.g., corrupted sensor data, adversarial input, systematic model error) cannot defeat both the primary output and its validation.*

[V.AI-5.5] INDEPENDENT OUTPUT VALIDATION

The AI system shall verify independent output validation by analysis and test.

The analysis shall identify independent validation sources for each safety-critical AI output and confirm validation sources use dissimilar methods, sensors, or processing from the primary AI. The analysis shall demonstrate that independent validation does not share common failure modes with the primary AI, including common training data sources, common sensor inputs, common processing hardware, or common software dependencies.

The test shall inject discrepancies between AI output and independent validation sources and confirm discrepancies trigger human review or defined fallback behavior, the system does not execute safety-critical actions when validation fails, and validation latency meets operational timing requirements.

Success Criteria: The verification shall be considered successful when the analysis and test show that independent validation sources are identified for each safety-critical AI output, validation sources use dissimilar methods from the primary AI,

discrepancies trigger human review or fallback behavior, and independent validation does not share common failure modes with the primary AI.

2.6 AI-6: OUT-OF-DISTRIBUTION DETECTION

[R.AI-6] OUT-OF-DISTRIBUTION DETECTION

The AI system shall detect and appropriately handle inputs that fall outside the distribution of training data.

ML models are only reliable within the distribution of data they were trained on. When inputs differ significantly from training data due to sensor anomalies, novel operational scenarios, or environmental conditions not anticipated during development model outputs become unreliable and potentially dangerous. Traditional software has defined input/output bounds that can be checked explicitly; AI systems are vulnerable to OOD inputs that may appear valid but produce unreliable predictions. This requirement addresses the “vulnerable to OOD inputs” gap identified in Section 1.2.4.2, ensuring the system recognizes when it is operating outside its validated envelope.

Applicable Domain Standards

- General: ISO/IEC 23053:2022; ISO/IEC 5338:2023 §6.4.14
- Aviation: FAA AI Safety Roadmap (operational envelope)
- Automotive: ISO 21448:2022 §6–§8; ISO 34503 (ODD boundary definitions)
- Medical: FDA AI/ML-Based SaMD Action Plan (real-world performance monitoring)
- EU: EU AI Act Article 15 (Robustness to errors)
- Cross-Domain: NIST AI RMF MEASURE-2.6

2.6.1 Training Distribution Characterization

[R.AI-6.1] TRAINING DISTRIBUTION CHARACTERIZATION

The AI system shall characterize the training data distribution for each ML model.

Rationale: OOD detection requires knowing what “*in-distribution*”^{*} looks like. Characterizing the training data distribution provides the baseline against which operational inputs are compared. Documentation should include statistical measures (mean, variance, range, density estimates) that enable automated OOD ^{**}detection at runtime. Without explicit distribution characterization, OOD detection cannot be implemented systematically. The distribution shall be

characterized within the scope declared under AI-1.0; characterization need not extend beyond the declared Operational Environment and Subject Population.*

[V.AI-6.1] TRAINING DISTRIBUTION CHARACTERIZATION

The AI system shall verify training distribution characterization by inspection.

The inspection shall confirm training data distribution is documented for each model, distribution boundaries and characteristics are defined including statistical measures (e.g., mean, variance, covariance, range, density estimates), documentation includes statistical measures of distribution properties sufficient to support runtime OOD detection, and distribution characterization is traceable to training datasets.

Success Criteria: The verification shall be considered successful when the inspection shows that training data distribution is documented for each model, distribution boundaries and characteristics are defined, statistical measures are documented, and characterization is traceable to training datasets.

2.6.2 Runtime OOD Detection

[R.AI-6.2] RUNTIME OOD DETECTION

The AI system shall implement mechanisms to detect out-of-distribution inputs at runtime.

Rationale: *OOD detection must occur during operation, not just during development testing. Runtime detection mechanisms compare incoming inputs against the characterized training distribution and flag inputs that fall outside defined boundaries. Detection methods may include statistical distance measures (Mahalanobis distance), density estimation, ensemble disagreement, or dedicated OOD detection models. Detection must meet operational timing requirements to enable timely response.*

[V.AI-6.2] RUNTIME OOD DETECTION

The AI system shall verify runtime OOD detection by test.

The test shall confirm OOD detection mechanisms are implemented for each ML model, detection correctly identifies inputs outside training distribution using defined statistical methods, detection performance is validated against known OOD test cases including both synthetic OOD inputs and realistic operational scenarios, and detection latency meets operational timing requirements for the AI function.

Success Criteria: The verification shall be considered successful when the test shows that OOD detection mechanisms are implemented for each model, detection

correctly identifies OOD inputs, detection is validated against known test cases, and latency meets operational timing requirements.

2.6.3 OOD Response

[R.AI-6.3] OOD RESPONSE

The AI system shall prevent autonomous action on out-of-distribution inputs without human review or defined fallback behavior.

Rationale: *Detecting OOD inputs without a defined response provides no protection. When an input is flagged as OOD, the system must either escalate to human review or invoke predefined fallback behavior appropriate to the AI function's criticality. Autonomous execution based on potentially unreliable OOD predictions could lead to hazardous outcomes. The response should be proportionate to system classification Safety-Critical AI may require mandatory human review, while Operational Support AI may use automated fallback.*

[V.AI-6.3] OOD RESPONSE

The AI system shall verify OOD response by test.

The test shall inject OOD inputs and confirm OOD inputs trigger defined response procedures, autonomous execution is blocked or fallback behavior is invoked, human review workflow is available for OOD conditions when required by system classification, and system behavior for OOD inputs is documented and tested across all defined OOD scenarios.

Success Criteria: The verification shall be considered successful when the test shows that OOD inputs trigger defined response procedures, autonomous execution is blocked or fallback is invoked, human review workflow is available when required, and OOD behavior is documented and tested.

2.6.4 OOD Event Logging

[R.AI-6.4] OOD EVENT LOGGING

The AI system shall log all out-of-distribution detection events.

Rationale: *OOD events provide valuable information about operational conditions that differ from training assumptions. Logging enables post-mission analysis to identify patterns, assess whether the operational environment has shifted from training assumptions, and inform decisions about model retraining or operational envelope expansion. OOD logs also support trend analysis to detect gradual distribution shift that might indicate data drift.*

[V.AI-6.4] OOD EVENT LOGGING

The AI system shall verify OOD event logging by inspection.

The inspection shall confirm OOD events are logged with timestamps and input data (or input characterization sufficient for analysis), logs capture detection confidence and method, logs capture system response and operator actions, and logs are retrievable for trend analysis and model improvement.

Success Criteria: The verification shall be considered successful when the inspection shows that OOD events are logged with timestamps and input data, logs capture detection confidence, method, and system response, and logs are retrievable for analysis.

2.7 AI-7: ADVERSARIAL ROBUSTNESS

[R.AI-7] ADVERSARIAL ROBUSTNESS

The AI system shall implement mechanisms to detect, prevent, and mitigate adversarial attacks that could compromise AI integrity, manipulate outputs, or degrade system performance.

AI systems are vulnerable to attack vectors that do not exist for traditional software. Adversarial attacks can manipulate training data (poisoning), craft inputs that cause misclassification (adversarial examples), extract model parameters (model theft), or reconstruct sensitive training data (model inversion). Adversarial compromise could affect safety or operational success. This requirement addresses the “vulnerable to adversarial attacks” gap identified in Section 1.2.4.

AI-specific adversarial robustness does not replace foundational cybersecurity engineering; it extends it. The controls in AI-7.1 through AI-7.6 address attack vectors unique to AI systems; however, these controls assume an underlying cybersecurity posture consistent with established security frameworks. Systems subject to SCL certification should implement security controls consistent with NIST SP 800-53 Rev. 5 (Security and Privacy Controls for Information Systems and Organizations), with particular attention to the System and Information Integrity (SI) and System and Communications Protection (SC) control families as they apply to AI components. Additionally, the development process for AI system software, including ML pipeline code, training infrastructure, and deployment tooling, should follow secure software development practices consistent with NIST SP 800-218 (Secure Software Development Framework, SSDF). AI-7 sub-requirements complement rather than duplicate these foundational controls: data poisoning protection (AI-7.1) extends SI controls to training data; model integrity (AI-7.4) extends SI controls to model artifacts; and supply chain security (AI-7.6) extends

SSDF practices to AI-specific dependencies including pre-trained models and ML frameworks.

Applicable Domain Standards

- General: NIST AI 100-2e2023 (Adversarial ML Taxonomy); ISO/IEC 5338:2023 §6.4.3
- Aviation: DO-326A (Airworthiness Security Process); DO-356A (Airworthiness Security Methods); FAA AI Safety Roadmap
- Automotive: ISO/SAE 21434:2021 (Road vehicles cybersecurity); UNECE WP.29 R155 (Cyber security management)
- Medical: FDA Premarket Cybersecurity Guidance (2023); FDA Postmarket Cybersecurity Guidance (2016)
- EU: EU AI Act Article 15 (Cybersecurity); EU Cyber Resilience Act [verify article]
- Cross-Domain: NIST AI RMF MANAGE-2.2; NIST SP 800-218 (Secure Software Development Framework); NIST SP 800-53 Rev. 5 SI family

2.7.1 Data Poisoning Protection

[R.AI-7.1] DATA POISONING PROTECTION

The AI system shall implement protections against training data poisoning attacks.

Rationale: Data poisoning injects malicious samples into training datasets to influence model behavior in attacker-controlled ways. Poisoned models may appear to function normally on most inputs but produce incorrect outputs on specific attacker-chosen inputs. Poisoning could create backdoors that trigger dangerous behavior under specific conditions. Protection requires authenticating data sources, validating data integrity, and detecting statistically anomalous samples.

[V.AI-7.1] DATA POISONING PROTECTION

The AI system shall verify data poisoning protection by analysis and test.

The analysis shall confirm training data sources are authenticated and validated through documented provenance, and data integrity checks detect unauthorized modifications.

The test shall confirm anomaly detection identifies statistically suspicious training samples, and poisoning attack scenarios are tested with known attack patterns (e.g., backdoor triggers, label flipping, gradient-based poisoning).

Success Criteria: The verification shall be considered successful when the analysis and test show that training data sources are authenticated, data integrity checks detect modifications, anomaly detection identifies suspicious samples, and poisoning attack scenarios are tested.

Applicable Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-7 (Software, firmware, and information integrity); ISO/IEC 27001:2022 A.8.28 (Secure coding)
- AI-Specific: NIST AI 100-2e2023 §3.2 (Data poisoning taxonomy); MITRE ATLAS AML.T0020 (Poison training data)
- Data Integrity: ISO/IEC 5259-3:2024 (Data quality management, verification); ISO 8000-61:2016 (Data quality management process reference model)
- Aviation: RTCA DO-326A §3.5 (Security risk assessment); SAE ARP 4754A §4 (Development assurance)
- Automotive: ISO/SAE 21434:2021 §8.3 (Cybersecurity risk assessment)
- Cross-Domain: NIST AI RMF MANAGE-2.1 (Risk response to identified AI risks); EU AI Act Article 15(4) (Resilience against attempts to manipulate training data)

2.7.2 Adversarial Input Protection

[R.AI-7.2] ADVERSARIAL INPUT PROTECTION

The AI system shall implement protections against adversarial input manipulation at runtime.

Rationale: *Adversarial inputs are carefully crafted to cause model misclassification while appearing valid to human observers. Small perturbations imperceptible to humans can cause dramatic changes in model outputs. Known attack patterns include FGSM (Fast Gradient Sign Method), PGD (Projected Gradient Descent), and C&W (Carlini & Wagner) attacks. Runtime protection requires input validation, perturbation detection, and verification that outputs are stable under small input changes.**

[V.AI-7.2] ADVERSARIAL INPUT PROTECTION

The AI system shall verify adversarial input protection by test.

The test shall confirm input validation detects malformed or anomalous inputs, adversarial perturbation detection mechanisms are implemented, model outputs are stable under small input perturbations within operational bounds, and known adversarial attack patterns (e.g., FGSM, PGD, C&W) are tested and model response is documented.

Success Criteria: The verification shall be considered successful when the test shows that input validation detects anomalous inputs, perturbation detection is implemented, outputs are stable under small perturbations, and known attack patterns are tested.

Applicable Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-10 (Information input validation); NIST AI 100-2e2023 §2 (Evasion attacks)
- AI-Specific: OWASP ML Top 10 (2023) MLO1 (Input manipulation attack); MITRE ATLAS AML.TOO43 (Craft adversarial data)
- Robustness: ISO/IEC TR 24029-1:2021 (AI robustness assessment overview); IEEE 7009-2024 (Fail-safe design of autonomous systems)
- Aviation: RTCA DO-326A §3.7 (Security architecture); SAE ARP 6983 (AI/ML in airborne systems, draft)
- Automotive: ISO/SAE 21434:2021 §9.4 (Cybersecurity concept)
- Cross-Domain: NIST AI RMF MEASURE-2.7 (AI system security); EU AI Act Article 15(5) (Resilience against adversarial inputs)

2.7.3 Model Inversion and Extraction Protection

[R.AI-7.3] MODEL INVERSION AND EXTRACTION PROTECTION

The AI system shall implement protections against model inversion and extraction attacks.

Rationale: *Model inversion attacks reconstruct sensitive training data by analyzing model outputs potentially exposing sensitive information. Model extraction attacks systematically query the model to replicate its behavior, enabling attackers to study vulnerabilities offline. Protection requires access controls, query rate limiting, and output perturbation techniques that prevent systematic probing while maintaining operational utility. Where differential privacy or analogous techniques are applied for data subject privacy purposes, see also AI-10.4 (Privacy-Enhancing Techniques); the same mechanism may satisfy both requirements under distinct threat models.*

[V.AI-7.3] MODEL INVERSION AND EXTRACTION PROTECTION

The AI system shall verify model inversion and extraction protection by inspection and analysis.

The inspection shall confirm model access is restricted to authorized queries only, and training data sensitivity classification is documented.

The analysis shall confirm query rate limiting prevents systematic model probing, and output perturbation or differential privacy techniques are applied where applicable based on training data sensitivity.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that model access is restricted, training data sensitivity is classified, query rate limiting is implemented, and appropriate privacy techniques are applied.

Applicable Domain Standards

- General: NIST SP 800-53 Rev. 5 SC-8 (Transmission confidentiality) and SC-28 (Protection of information at rest)
- AI-Specific: NIST AI 100-2e2023 §5 (Privacy attacks — membership inference, model inversion, extraction); MITRE ATLAS AML.TOO24 (Exfiltration via ML inference API)
- Privacy-Preserving ML: NIST SP 800-226 (Guidelines for evaluating differential privacy guarantees)
- Data Protection: ISO/IEC 27701:2019 (Privacy information management); GDPR Article 32 (Security of processing)
- Cross-Domain: NIST AI RMF MAP-4.1 (Approaches for mapping AI technology and legal risks); EU AI Act Article 15(4) (Resilience against confidentiality attacks)

2.7.4 Model Integrity

[R.AI-7.4] MODEL INTEGRITY

The AI system shall maintain model integrity throughout the operational lifecycle.

Rationale: Model integrity ensures that the deployed model is the validated model and has not been modified, corrupted, or replaced. Unauthorized modifications could introduce backdoors, degrade performance, or alter behavior in ways that circumvent all prior verification. Cryptographic signing provides tamper-evident verification, and integrity checks at load time and runtime detect unauthorized

changes. Model integrity controls apply in concert with the maintenance criteria established under AI-4.3 and the deployment format validation established in Section 3.3; legitimate model updates shall pass through all three control points before operational deployment.

[V.AI-7.4] MODEL INTEGRITY

The AI system shall verify model integrity by inspection.

The inspection shall confirm model artifacts are cryptographically signed, integrity verification occurs at model load and at defined intervals during operation, unauthorized model modifications trigger alerts and fallback procedures, and model version history is maintained with tamper-evident logging.

Success Criteria: The verification shall be considered successful when the inspection shows that model artifacts are signed, integrity verification occurs at load and defined intervals, unauthorized modifications trigger alerts, and version history is maintained with tamper-evident logging.

Applicable Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-7 (Integrity) and CM-5 (Access restrictions for change)
- Secure Development: NIST SP 800-218 (SSDF — Secure Software Development Framework); SLSA Framework v1.0 (Supply-chain Levels for Software Artifacts)
- Model Signing: Sigstore / Linux Foundation Model Transparency project (cryptographic signing of model weights)
- Aviation: RTCA DO-326A §3.9 (Security assurance); RTCA DO-178C §11.5 (Configuration management)
- Automotive: ISO/SAE 21434:2021 §8.8 (Cybersecurity validation)
- Cross-Domain: NIST AI RMF MANAGE-2.1 (Risk response); EU AI Act Article 15(1) (Accuracy, robustness, and cybersecurity throughout lifecycle)

2.7.5 Adversarial Event Logging

[R.AI-7.5] ADVERSARIAL EVENT LOGGING

The AI system shall log all adversarial detection events and security incidents for post-mission analysis.

Rationale: *Comprehensive logging of adversarial events supports post-mission forensic analysis, enables detection of attack patterns, and informs security improvements for future missions. Logs must be immutable to prevent attacker manipulation and must be retained for sufficient duration to support post-mission analysis.*

[V.AI-7.5] ADVERSARIAL EVENT LOGGING

The AI system shall verify adversarial event logging by inspection.

The inspection shall confirm security events are logged with timestamps, event type, and severity, logs capture detection confidence and system response, logs are immutable and retrievable for forensic analysis, and log retention meets mission duration plus post-mission analysis requirements.

Success Criteria: The verification shall be considered successful when the inspection shows that security events are logged with timestamps, type, and severity, logs capture detection confidence and response, logs are immutable and retrievable, and retention meets requirements.

Applicable Domain Standards

- General: NIST SP 800-53 Rev. 5 AU-2 (Event logging), AU-3 (Content of audit records), AU-12 (Audit record generation)
- Security Monitoring: ISO/IEC 27001:2022 A.8.15 (Logging) and A.8.16 (Monitoring activities)
- Incident Response: NIST SP 800-61 Rev. 2 (Computer security incident handling guide)
- Aviation: RTCA DO-326A §3.10 (Security monitoring)
- Automotive: ISO/SAE 21434:2021 §10.3 (Cybersecurity monitoring operations)
- Cross-Domain: NIST AI RMF MEASURE-4.3 (Ongoing monitoring); EU AI Act Article 12 (Record-keeping of events relevant to risk identification)

2.7.6 AI Supply Chain Security

[R.AI-7.6] AI SUPPLY CHAIN SECURITY

The AI system shall assess and document supply chain risks for all AI/ML frameworks, libraries, and pre-trained model components.

Rationale: *AI systems depend on complex software supply chains including ML frameworks (e.g., TensorFlow, PyTorch), libraries (e.g., NumPy, ONNX), and potentially

pre-trained models or transfer learning components. These dependencies may contain vulnerabilities, backdoors, or undocumented behaviors introduced through foreign contribution, compromised maintainers, or malicious updates. Pre-trained models may encode biases, backdoors, or behaviors from unknown training data. Supply chain assessment ensures that AI system integrity extends beyond the project's own development to encompass all external dependencies.*

[V.AI-7.6] AI SUPPLY CHAIN SECURITY

The AI system shall verify AI supply chain security by inspection.

The inspection shall confirm AI/ML frameworks and libraries are identified and documented with version numbers and sources, supply chain risk assessment addresses foreign contribution, maintenance status, and known vulnerabilities, pre-trained models (if used) have documented provenance and integrity verification, and mitigation strategies are documented for identified supply chain risks.

Success Criteria: The verification shall be considered successful when the inspection shows that AI/ML frameworks and libraries are documented, supply chain risk assessment addresses contribution sources and maintenance, pre-trained model provenance is documented, and mitigation strategies are documented for identified risks.

Applicable Domain Standards

- General: NIST SP 800-53 Rev. 5 SR-3 (Supply chain controls) and SR-5 (Acquisition strategies)
- Supply Chain: NIST SP 800-161 Rev. 1 (C-SCRM for systems and organizations); NIST SP 800-218 (SSDF)
- Model Bill-of-Materials: NTIA Minimum Elements for SBOM (2021); CISA SBOM guidance — extensions for ML (ML-BOM)
- Aviation: RTCA DO-326A §3.8 (Supplier risk); RTCA DO-355 (Information security guidance for airworthiness)
- Automotive: ISO/SAE 21434:2021 §15 (Distributed cybersecurity activities — supplier relationships)
- Cross-Domain: NIST AI RMF GOVERN-6.1 (Third-party risk management); EU AI Act Article 25 (Obligations along the AI value chain)

2.8 AI-8: EXPLAINABILITY

[R.AI-8] EXPLAINABILITY

The AI system shall provide explanations of AI decisions and recommendations sufficient for operators to verify AI reasoning and establish appropriate trust.

Traditional software has explicit decision logic that can be inspected through code review an engineer can trace exactly why a particular output was produced. AI systems, particularly neural networks, make decisions through learned patterns in weights that are not directly interpretable. ISO 22989 Section 3.5.7 defines explainability as the property of an AI system to express important factors influencing results in a way humans can understand. Without explainability, operators cannot verify AI reasoning, leading to either blind over-trust or unwarranted under-trust. This requirement addresses the “black-box decision making” gap identified in Section 1.2.4.2. Where AI-10.3 (Data Subject Rights) requires explanation of automated decisions, AI-8 explainability outputs serve as supporting evidence.

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §3.5.7; ISO/IEC TS 6254:2024 (AI system explainability) [verify]
- Aviation: FAA AI Safety Roadmap (interpretability requirements)
- Medical: FDA Transparency Principles for Medical Devices with AI/ML (2021); FDA CDS Guidance (2022) §V.A
- EU: EU AI Act Article 13 (Transparency and provision of information)
- Cross-Domain: NIST AI RMF MANAGE-4.3; NIST SP 1270 (explainability and bias intersection)

2.8.1 Explainability Requirements Definition

[R.AI-8.1] EXPLAINABILITY REQUIREMENTS DEFINITION

The AI system shall define explainability requirements appropriate to each AI function’s criticality and user needs.

Rationale: *Explainability requirements should scale with decision criticality Safety-Critical AI functions require more rigorous explanation capabilities than Operational Support functions. Requirements must also consider the target audience; operators need operationally relevant explanations, while engineers may need deeper technical detail. Defining requirements upfront ensures explainability is designed in, not retrofitted.*

[V.AI-8.1] EXPLAINABILITY REQUIREMENTS DEFINITION

The AI system shall verify explainability requirements definition by inspection.

The inspection shall confirm explainability requirements are defined for each AI function, requirements are scaled to function criticality and decision impact, target audience and their explanation needs are documented (e.g., operators, engineers, analysts), and explainability approach is justified for each AI function (e.g., feature importance, attention visualization, rule extraction, inherently interpretable models).

Success Criteria: The verification shall be considered successful when the inspection shows that explainability requirements are defined for each AI function, requirements scale with criticality, target audiences are documented, and explainability approaches are justified.

2.8.2 Decision Explanation Generation

[R.AI-8.2] DECISION EXPLANATION GENERATION

The AI system shall provide decision explanations in a format understandable to operators.

Rationale: *Explanations must be actionable and comprehensible to the intended audience within operational time constraints. Technical explanations (e.g., feature weights, activation patterns) may be appropriate for post-mission analysis but are inadequate for real-time operational decisions. Explanation format should be validated through human factors assessment to ensure operators actually understand and can act on the information provided.*

[V.AI-8.2] DECISION EXPLANATION GENERATION

The AI system shall verify decision explanation generation by test.

The test shall confirm explanations are generated for AI decisions and recommendations, explanation format is appropriate to operational context and timing constraints, explanations identify key factors influencing the decision, and operator comprehension is validated through usability assessment with representative operators performing representative tasks.

Success Criteria: The verification shall be considered successful when the test shows that explanations are generated for AI decisions, format is appropriate to context, key influencing factors are identified, and operator comprehension is validated through usability assessment.

2.8.3 Confidence Indication

[R.AI-8.3] CONFIDENCE INDICATION

The AI system shall indicate confidence levels for AI outputs.

Rationale: *AI outputs vary in reliability some predictions are made with high confidence, others are uncertain. Operators need to know when to trust AI outputs and when to apply additional scrutiny or seek alternative information. Confidence must be calibrated a stated 90% confidence should be correct approximately 90% of the time to enable appropriate operator trust calibration. ISO 22989 Section 3.5.8 defines predictability as enabling reliable assumptions about outputs; calibrated confidence supports this property.*

[V.AI-8.3] CONFIDENCE INDICATION

The AI system shall verify confidence indication by test.

The test shall confirm confidence metrics are computed for AI outputs, confidence is presented to operators with outputs in a format that supports decision-making, confidence calibration is validated against actual accuracy (e.g., Expected Calibration Error ≤ 0.05), and low-confidence outputs are clearly distinguished from high-confidence outputs through visual or auditory differentiation.

Success Criteria: The verification shall be considered successful when the test shows that confidence metrics are computed, confidence is presented with outputs, calibration is validated against actual accuracy, and low-confidence outputs are clearly distinguished.

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §3.5.8 (Predictability); ISO/IEC TS 6254:2024 §6 (Confidence communication) [verify section]
- Aviation: FAA AI Safety Roadmap §3 (Interpretability and uncertainty quantification) [verify section]
- Automotive: SAE J3016:2021 §5.3 (Automation reliability feedback to user)
- Medical: FDA AI/ML Transparency Principles (2021) Principle 3 (Labeling for transparency); FDA CDS Guidance (2022) §V.B (Evidence basis and strength of evidence)
- EU: EU AI Act Article 13(3)(b)(iv) (Information on level of accuracy and reliability)
- Cross-Domain: NIST AI RMF MEASURE-2.5

2.8.4 Reasoning Inspection Capability

[R.AI-8.4] REASONING INSPECTION CAPABILITY

The AI system shall enable operators to query or inspect AI reasoning for safety-critical decisions.

Rationale: *For safety-critical decisions, operators may need to understand AI reasoning in greater depth than routine explanations provide. On-demand inspection capability allows operators to drill down into AI reasoning when uncertainty exists or when AI recommendations conflict with operator **expectations. This capability supports the human decision authority required by AI-9 and aligns with ISO 22989 Section 3.5.6 controllability requirements.*

[V.AI-8.4] REASONING INSPECTION CAPABILITY

The AI system shall verify reasoning inspection capability by test.

The test shall confirm query or inspection capability is available for safety-critical AI functions, operators can access additional detail on demand beyond routine explanations, inspection does not unacceptably delay operational timelines, and inspection capability is documented in operator procedures.

Success Criteria: The verification shall be considered successful when the test shows that inspection capability is available for safety-critical functions, operators can access additional detail on demand, inspection meets timing requirements, and capability is documented in procedures.

2.8.5 Public Disclosure Support

[R.AI-8.5] PUBLIC DISCLOSURE SUPPORT

The AI system shall provide the artifacts and information necessary to support public disclosure obligations applicable to the deployment context, in a form suitable for stakeholder consumption outside the development organization.

Rationale: AI systems may be subject to public disclosure requirements including external model cards or system factsheets, consumer-facing information under EU AI Act Article 13, federal AI use case inventory reporting under Executive Order 14110 and OMB M-24-10, and public registration for high-risk systems under EU AI Act Article 71. The specific disclosure obligations depend on deployment context and generally bind the deploying organization rather than the developer; however, the AI system design must support those obligations by producing standardized disclosure artifacts consumable by regulators, users, data subjects, and the public. This requirement complements the internal explainability requirements (AI-8.1 through AI-8.4) by establishing the external-facing disclosure surface. Disclosure artifacts should be derived from existing framework evidence (ODD declaration per AI-1.0, bias evaluation per AI-2, performance characterization per AI-4, explainability outputs per AI-8.1-8.4) rather than generated independently, to ensure external

disclosures remain consistent with internal verification evidence. This sub-requirement is optional by classification and may be documented as Not Applicable with rationale where no public disclosure obligation applies.

[V.AI-8.5] PUBLIC DISCLOSURE SUPPORT

The AI system shall verify public disclosure support by inspection.

The inspection shall confirm that applicable public disclosure obligations are identified in the ODD declaration under Regulatory and Operational Authorization (AI-1.O); that disclosure artifact formats are defined (e.g., external model card, AI factsheet, EU AI Act Article 13 user information, federal AI inventory entry); that each required artifact is populated from existing framework evidence rather than generated in parallel; and that artifacts are reviewed prior to release for consistency with ODD declaration, documented limitations, and verification evidence.

The inspection shall further confirm that artifact update procedures are defined so that disclosure artifacts remain consistent with the current system state following updates, drift remediation (AI-6.4), reclassification (AI-9.6), or ODD revision; and that artifacts are retained for the retention period required by the applicable regulatory regime.

Success Criteria: The verification shall be considered successful when the inspection shows that applicable disclosure obligations are identified, disclosure artifacts are defined and populated from framework evidence, artifacts are reviewed for consistency before release, update procedures are defined, and retention requirements are satisfied. Where no public disclosure obligation applies, AI-8.5 may be documented as Not Applicable with rationale.

Applicable Domain Standards

- General: ISO/IEC 23894:2023 §6.4.5 (Transparency and explainability to stakeholders); ISO/IEC 5259-4 §7 (Data quality reporting); ISO/IEC 42001:2023 §8.3 (Communication)
- Aviation: FAA AI Safety Roadmap (stakeholder transparency)
- Automotive: NHTSA AV 4.0 element 2 (Operational Design Domain description); NHTSA AV Transparency and Engagement Principles
- Medical: FDA Predetermined Change Control Plan Guidance (2024) §VI (Device labeling); FDA Transparency for Machine Learning-Enabled Device Guidance (2021)

- EU: EU AI Act Article 13 (Transparency and provision of information to deployers); EU AI Act Article 50 (Transparency obligations for general-purpose AI); EU AI Act Article 71 (Public EU database registration for high-risk AI)
- US Federal: Executive Order 14110 Section 10.1(b) (Federal AI inventory); OMB M-24-10 (Federal AI use case inventory requirements); NIST AI 100-1 (AI Risk Management Framework)
- Cross-Domain: NIST AI RMF GOVERN-1.4 (Risk documentation); NIST AI RMF MANAGE-4.1 (Post-deployment monitoring disclosure)

2.9 AI-9: HUMAN-AI TEAMING

[R.AI-9] HUMAN-AI TEAMING

The AI system shall support effective human-AI collaboration, enabling operators to maintain situational awareness, appropriate trust calibration, and final decision authority over safety-critical actions.

AI systems in safety-critical operations operate as part of a human-machine team, not as fully autonomous agents. ISO 22989 Section 5.13 defines automation levels ranging from full human control to full autonomy; for safety-critical systems, human oversight remains essential. Operators must understand what the AI is doing, trust it appropriately (neither too much nor too little), and retain authority to override AI actions when necessary. This requirement addresses the “operators cannot verify reasoning” gap and ensures AI augments rather than replaces human judgment in safety-critical decisions.

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §3.5.6 (Controllability); §5.13 (Automation levels)
- Aviation: 14 CFR 25.1302 (Installed systems and equipment for use by the flightcrew); FAA AC 25.1322 (Flight Crew Alerting); FAA AI Safety Roadmap (human oversight)
- Automotive: SAE J3016:2021 §5 (Automation levels); NHTSA Automated Vehicles Comprehensive Plan (2021) [verify section]
- Medical: FDA CDS Guidance (2022) §V (Practitioner review); HHS/ONC Health IT Certification

- EU: EU AI Act Article 14 (Human oversight); EU AI Act Article 26 (Deployer obligations)
- Cross-Domain: NIST AI RMF GOVERN-1.5; NIST AI RMF MANAGE-2.1

2.9.1 Human Decision Authority

[R.AI-9.1] HUMAN DECISION AUTHORITY

The AI system shall maintain human decision authority over safety-critical actions.

Rationale: *For safety-critical actions those that could result in harm, equipment damage, or mission loss humans must retain final decision authority. AI may recommend, advise, or prepare actions, but execution should require human authorization except where explicitly designed and approved for autonomous operation (e.g., time-critical emergency responses). ISO 22989 Section 3.5.6 **defines controllability as the property allowing human intervention; this requirement ensures controllability is maintained for safety-critical functions.*

[V.AI-9.1] HUMAN DECISION AUTHORITY

The AI system shall verify human decision authority by test.

The test shall confirm safety-critical actions require human authorization before execution, AI cannot execute safety-critical actions autonomously unless explicitly designed, approved, and documented for autonomous operation, human override capability is always available and functional, and authority boundaries are documented and enforced through technical controls.

Success Criteria: The verification shall be considered successful when the test shows that safety-critical actions require human authorization, autonomous execution is limited to explicitly approved cases, human override is always available, and authority boundaries are documented and enforced.

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §3.5.6 (Controllability)
- Aviation: 14 CFR 25.1302 (Flight crew installed systems); FAA AC 25.1322 §4 (Alert prioritization and integration)
- Automotive: SAE J3016:2021 §5.2 (Dynamic Driving Task fallback); NHTSA AV 4.0 Voluntary Safety Self-Assessment element 3 (Human Machine Interface)
- Medical: FDA Clinical Decision Support Software Guidance (2022) §V.A (Basis for practitioner independent review)

- EU: EU AI Act Article 14(4) (Human oversight measures for high-risk AI); EU AI Act Article 14(5) (oversight of certain biometric categorization)
- Cross-Domain: NIST AI RMF GOVERN-1.5; NIST AI RMF MANAGE-2.1

2.9.2 Operational Situational Awareness

[R.AI-9.2] OPERATIONAL SITUATIONAL AWARENESS

The AI system shall support operator situational awareness of AI state and behavior.

Rationale: Operators cannot effectively supervise AI systems they do not understand. Situational awareness requires visibility into what the AI is doing, what it is recommending, and why. Changes in AI state (e.g., mode transitions, confidence drops, detected anomalies) must be communicated to operators to prevent surprise and enable proactive response.

[V.AI-9.2] OPERATIONAL SITUATIONAL AWARENESS

The AI system shall verify operator situational awareness support by test.

The test shall confirm AI operational state is visible to operators through appropriate displays, AI actions and recommendations are communicated clearly within operational timing constraints, changes in AI state or confidence are annunciated (e.g., state change notification $\leq 500\text{ms}$, confidence drop $>15\%$ triggers alert), and operators can determine what the AI is doing and why through available displays and explanations.

Success Criteria: The verification shall be considered successful when the test shows that AI state is visible to operators, actions and recommendations are clearly communicated, state changes are annunciated within timing requirements, and operators can determine AI status and reasoning.

2.9.3 Trust Calibration

[R.AI-9.3] TRUST CALIBRATION

The AI system shall support appropriate trust calibration between operators and AI.

Rationale: Miscalibrated trust either over-trust or under-trust degrades human-AI team performance. Over-trust leads to automation complacency where operators fail to catch AI errors; under-trust leads to operators ignoring valid AI recommendations. Trust calibration requires that operators understand AI reliability and limitations, that training addresses appropriate trust levels, and that system design promotes calibrated trust through transparency and appropriate confidence communication.

[V.AI-9.3] TRUST CALIBRATION

The AI system shall verify trust calibration support by analysis and test.

The analysis shall confirm AI reliability and limitations are documented and communicated to operators, and training materials address appropriate AI trust levels including scenarios where AI should and should not be trusted.

The test shall confirm system design discourages both over-trust (e.g., through uncertainty indication, limitation warnings) and under-trust (e.g., through reliability demonstration, confidence calibration), and trust calibration is assessed through human factors evaluation with representative operators.

Success Criteria: The verification shall be considered successful when the analysis and test show that AI reliability and limitations are communicated, training addresses trust calibration, system design discourages miscalibrated trust, and human factors evaluation confirms appropriate trust calibration.

2.9.4 Graceful Degradation

[R.AI-9.4] GRACEFUL DEGRADATION

The AI system shall degrade gracefully when AI capabilities are reduced or unavailable.

Rationale: *AI systems may experience capability reduction due to sensor failures, computational resource constraints, detected anomalies, or explicit mode transitions. When AI capabilities are degraded, operators must be notified and manual or backup procedures must be available to continue operations. Abrupt AI failure without graceful degradation could leave operators without situational awareness or control capability. Degraded mode definitions shall be consistent with the System Inputs degradation behaviors declared under AI-1.0.*

[V.AI-9.4] GRACEFUL DEGRADATION

The AI system shall verify graceful degradation by test.

The test shall confirm degraded AI modes are defined and documented for each AI function, operators are notified of AI capability reduction through appropriate annunciation, manual or backup procedures are available and documented for operation without full AI capability, and transition to degraded modes is tested and acceptable (e.g., ≤ 2 operator actions, ≤ 5 seconds to transition to manual mode).

Success Criteria: The verification shall be considered successful when the test shows that degraded modes are defined, operators are notified of capability

reduction, manual/backup procedures are available, and mode transitions are tested and acceptable.

2.9.5 Human-AI Interaction Logging

[R.AI-9.5] HUMAN-AI INTERACTION LOGGING

The AI system shall log human-AI interactions for post-mission analysis.

Rationale: *Comprehensive logging of human-AI interactions supports post-mission analysis to identify human factors issues, assess trust calibration, evaluate override patterns, and improve future AI systems. Logs should capture what the AI recommended, what the human decided, and the rationale for any overrides. This data is essential for continuous improvement of human-AI teaming effectiveness.*

[V.AI-9.5] HUMAN-AI INTERACTION LOGGING

The AI system shall verify human-AI interaction logging by inspection.

The inspection shall confirm human-AI interactions are logged with timestamps, logs capture AI recommendations and human decisions, logs capture overrides with rationale, and logs support post-mission human factors analysis including trust calibration assessment and override pattern analysis.

Success Criteria: The verification shall be considered successful when the inspection shows that interactions are logged with timestamps, logs capture recommendations, decisions, and override rationale, and logs support post-mission human factors analysis.

2.9.6 Operational Role Verification

[R.AI-9.6] OPERATIONAL ROLE VERIFICATION

The AI system shall implement periodic verification that the system's operational role remains consistent with its classification.

Rationale: AI system classification (Section 1.2.2) determines which requirements apply and at what stringency. Classification is assigned based on the intended operational role of the AI system: whether its outputs are advisory, operational, or safety critical. However, operational usage can drift over time without any change to the system itself. A system classified as Operational Support may produce outputs that operators gradually elevate from advisory to authoritative, relying on AI recommendations as though they were directives. This usage drift is particularly insidious because no technical change triggers a review. The model performs identically; only the human reliance on it has changed. When this occurs, the system is effectively operating at a higher classification tier without the corresponding

requirements, verifications, or safeguards. Periodic verification of operational role ensures that classification remains aligned with actual usage and that any drift toward higher reliance triggers formal reclassification and the application of additional requirements. This is the human factors complement to the technical drift monitoring required by AI-4.

[V.AI-9.6] OPERATIONAL ROLE VERIFICATION

The AI system shall verify operational role verification by inspection and analysis.

The inspection shall confirm a periodic review schedule is defined and justified based on system classification and operational tempo, the review process includes assessment of how operators actually use AI outputs (not just how the system is designed to be used), review criteria explicitly address whether operators treat AI outputs as advisory, operational, or authoritative, review criteria include adherence of actual operational use to the declared ODD (AI-1.O), and review findings are documented and retained as verification evidence.

The analysis shall confirm the review process includes mechanisms to detect usage drift, such as operator workflow analysis, override rate trends, or decision dependency assessment; that identified usage drift triggers formal reclassification evaluation per Section 1.2.2; and that when reclassification is warranted, the additional requirements corresponding to the new classification are applied and verified before continued operation at the elevated role.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that a periodic review schedule is defined and justified, the review assesses actual operator usage against classification assumptions, usage drift detection mechanisms are defined, identified drift triggers reclassification evaluation, and reclassification results in application of corresponding requirements.

2.9.7 Operator Qualification

[R.AI-9.7] OPERATOR QUALIFICATION

The AI system shall require documented operator qualifications appropriate to system classification and operational role, including system-specific training, demonstrated proficiency in normal and abnormal operations, and currency requirements that remain valid during continued operation.

Rationale: AI system safety and performance depend on operator understanding of system capabilities, limitations, and failure modes. Unqualified operators may over-trust AI outputs, fail to recognize degraded performance, or misinterpret advisory outputs as directives. Qualification rigor should scale with classification: Safety-

Critical AI requires formal operator certification, Mission-Critical AI requires role-based qualification, and Operational Support AI requires documented familiarization. Currency requirements prevent qualification decay during extended operation and ensure operators remain proficient following system updates that materially change AI behavior.

[V.AI-9.7] OPERATOR QUALIFICATION

The AI system shall verify operator qualification by inspection and test.

The inspection shall confirm that operator qualification requirements are documented for each operational role; that qualification criteria address AI-specific topics including system capabilities, documented limitations (Section 1.5 thresholds, AI-1.0 ODD bounds), known failure modes (AI-3 OOD, AI-7 adversarial), and appropriate trust calibration (AI-9.3); that currency requirements are defined with intervals justified by system classification and operational tempo; and that qualification records are retained as verification evidence.

The test shall confirm that operators assigned to the AI system hold current qualification for their role, that operators without current qualification are prevented from assuming roles that require it through administrative or technical controls, and that qualification re-validation is triggered when material system updates affect operator-relevant behavior.

Success Criteria: The verification shall be considered successful when the inspection and test show that operator qualification requirements are documented by role and classification, AI-specific qualification content is addressed, currency requirements are defined and enforced, and qualification records are retained as verification evidence.

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §3.5.5 (Human-AI collaboration); ISO/IEC 25059:2023 §5 (Quality characteristics for AI systems)
- Aviation: 14 CFR 121 Subpart N (Training Program); 14 CFR 121 Subpart Y (Advanced Qualification Program); FAA AI Safety Roadmap (operator training)
- Automotive: SAE J3016:2021 §8 (User considerations); NHTSA AV 4.0 element 12 (Consumer Education and Training)
- Medical: FDA 21 CFR 820.25 (Personnel – Training); FDA Clinical Decision Support Software Guidance (2022) §VI (Practitioner learning)

- EU: EU AI Act Article 4 (AI literacy); EU AI Act Article 14(4)(a) (Oversight competence); EU AI Act Article 26(2) (Deployer training obligations)
- Cross-Domain: NIST AI RMF GOVERN-2.2 (Accountability for staff); NIST SP 800-53 Rev. 5 AT-3 (Role-Based Training); NIST SP 800-53 Rev. 5 PS-3 (Personnel Screening)

2.9.8 Workload Management

[R.AI-9.8] WORKLOAD MANAGEMENT

The AI system shall define operational workload parameters and implement design measures that maintain operator cognitive capacity within documented limits under nominal, degraded, and high-tempo conditions.

Rationale: AI deployment can increase or decrease operator cognitive workload depending on interface design, alert logic, and the degree to which AI outputs require operator verification. Excess workload degrades decision quality, delays responses to abnormal conditions, and increases error probability. Insufficient workload induces complacency and reduces attentional resources available for monitoring AI outputs, which is a documented precursor to automation-induced failures. Workload must be validated as part of the operational design rather than discovered in deployment; this requirement complements trust calibration (AI-9.3) by ensuring the operator retains the cognitive margin required to exercise judgment and complements operator qualification (AI-9.7) by establishing the environment in which qualified operators function.

[V.AI-9.8] WORKLOAD MANAGEMENT

The AI system shall verify workload management by analysis and test.

The analysis shall confirm that operational workload parameters are defined for representative nominal, degraded, and high-tempo scenarios; that cognitive workload limits are documented with reference to an accepted workload assessment method (e.g., NASA-TLX, Bedford, SWAT) or domain guidance (e.g., FAA AC 25.1302, AAMI HE75); that automation-bias and complacency risks are explicitly addressed in the workload design; and that workload monitoring mechanisms (manual override rates, alert acknowledgement latency, task completion times) are specified.

The test shall confirm that under representative operational scenarios, measured workload remains within documented limits; that mitigations are invoked when workload indicators exceed defined thresholds (e.g., alert suppression, task offloading, fallback behaviors); and that operators retain the cognitive capacity necessary to exercise decision authority required by AI-9.1.

Success Criteria: The verification shall be considered successful when the analysis and test show that workload parameters are defined for nominal, degraded, and high-tempo scenarios, measured workload remains within documented limits, automation-bias and complacency risks are addressed, and workload mitigations are effective.

Applicable Domain Standards

- General: ISO 9241-11 (Usability: Definitions and concepts); ISO 11064 (Ergonomic design of control centres); ISO/IEC 25059:2023 §5 (AI system quality characteristics)
- Aviation: 14 CFR 25.1302 (Flightcrew workload); FAA AC 25.1302-1 (Installed Systems and Equipment for Use by the Flightcrew); FAA AI Safety Roadmap (cognitive workload)
- Automotive: NHTSA Visual-Manual Driver Distraction Guidelines (2013); SAE J3016:2021 §5.3 (Driver readiness); SAE J2944 (Driving performance terms)
- Medical: FDA Human Factors and Usability Engineering Guidance (2016) §5 (Task analysis); AAMI HE75:2009/(R)2018 (Human factors design)
- EU: EU AI Act Article 14(4)(d) (Measures to counter automation bias); Council Directive 89/391/EEC Article 6 (General obligations for safe working conditions)
- Cross-Domain: NIST AI RMF MEASURE-2.8 (Human factors); MIL-STD-1472H (DoD Human Engineering Design Criteria); NUREG-0711 Rev. 3 (Human Factors Engineering Program Review)

2.9.9 Training Program Requirements

[R.AI-9.9] TRAINING PROGRAM REQUIREMENTS

The AI system shall be supported by a training program that provides initial qualification, recurrent training, and scenario-based exercises covering AI system capabilities, documented limitations, failure modes, and appropriate trust calibration.

Rationale: Training programs for AI systems must address considerations that differ from traditional system training: non-deterministic behavior, performance variability across operational conditions, AI-specific failure modes such as OOD inputs and adversarial attacks, and the calibration between operator reliance and demonstrated AI performance. Initial qualification alone is insufficient because AI system behavior may evolve with updates, data drift, or operational context.

Recurrent training maintains proficiency during extended operation, and scenario-based exercises expose operators to rare failure modes before they occur operationally. Training program requirements complement operator qualification (AI-9.7) by defining the content and cadence that produce qualified operators, and complement workload management (AI-9.8) by ensuring operators possess the knowledge required to recognize and respond to high-workload conditions.

[V.AI-9.9] TRAINING PROGRAM REQUIREMENTS

The AI system shall verify training program requirements by inspection.

The inspection shall confirm that an initial qualification curriculum is documented and covers AI system capabilities, documented limitations (Section 1.5 thresholds, AI-1.0 ODD bounds), known failure modes (AI-3 OOD, AI-7 adversarial, AI-6 drift), explainability outputs (AI-8), and trust calibration guidance (AI-9.3); that recurrent training is scheduled at documented intervals justified by classification and operational tempo; that scenario-based exercises cover both nominal operations and rare off-nominal conditions including AI-specific failure modes; and that training effectiveness is measured through assessment of operator performance rather than attendance alone.

The inspection shall further confirm that material system updates trigger targeted training for the affected operator population before the updated system is released to operational use; that training content is updated to reflect current system behavior, thresholds, and ODD; and that training records are retained as verification evidence supporting AI-9.7 qualification currency.

Success Criteria: The verification shall be considered successful when the inspection shows that initial qualification, recurrent training, and scenario-based exercises are documented and delivered; that AI-specific content is addressed; that training is updated in response to material system changes; and that training records support AI-9.7 qualification currency.

Applicable Domain Standards

- General: ISO/IEC 22989:2022 §5.5 (Operation and monitoring); ISO/IEC 27002:2022 §6.3 (Information security awareness, education and training)
- Aviation: 14 CFR 121 Subpart Y (Advanced Qualification Program); FAA AC 120-35D (Line Operational Simulations); FAA AI Safety Roadmap §4 (Operator proficiency)
- Automotive: NHTSA AV 4.0 element 12 (Consumer education); SAE J3016:2021 §8.3 (User training for higher levels of driving automation)

- Medical: FDA 21 CFR 820.25 (Personnel training); FDA Clinical Decision Support Software Guidance (2022) §VI (Practitioner learning and proficiency)
- EU: EU AI Act Article 4 (AI literacy); EU AI Act Article 26(2) (Deployer obligations, competence of natural persons assigned oversight)
- Cross-Domain: NIST AI RMF GOVERN-2.2 (Accountability for staff); NIST SP 800-53 Rev. 5 AT-2 (Literacy Training), AT-3 (Role-Based Training), AT-4 (Training Records)

2.10 AI-10: PRIVACY AND DATA PROTECTION

[R.AI-10] PRIVACY AND DATA PROTECTION

The AI system will implement personal data protection mechanisms that enable lawful processing, respect data subject rights, and minimize privacy risk throughout the AI lifecycle.

AI systems frequently process personal data patient records in medical AI, biometric data in identity systems, behavioral data in recommendation systems, sensor-derived personal information in autonomous platforms. The handling of this data is governed by privacy regulations distinct from information classification regimes: GDPR in the EU, HIPAA in US healthcare, CCPA/CPRA in California, PIPEDA in Canada, and analogous frameworks globally. AI systems also introduce privacy risks not present in traditional software: training data may be reconstructable from model parameters (model inversion), trained models may leak membership information about training subjects, and the opacity of AI decisions complicates the exercise of data subject rights. This requirement addresses personal data handling throughout the AI lifecycle, complementing the information classification handling established in AI-1.5 through AI-1.8.

Applicability: AI-10 applies to AI systems that process personal data as defined under any applicable privacy framework. Where the system does not process personal data, AI-10 sub-requirements may be documented as Not Applicable with rationale. Determination of personal data processing shall be documented in the ODD declaration (AI-1.0) under Subject Population and Regulatory and Operational Authorization.

Applicable Domain Standards

- General: ISO/IEC 27701:2019 (Privacy Information Management); ISO/IEC 29100:2011 (Privacy framework)
- Medical: HIPAA Privacy Rule (45 CFR 164 Subpart E); HIPAA Security Rule (45 CFR 164 Subpart C)

- EU: GDPR (Regulation 2016/679); EU AI Act Article 10(5) (special categories of personal data)
- US State: CCPA/CPRA (Cal. Civ. Code §§ 1798.100 et seq.)
- Cross-Domain: NIST Privacy Framework v1.0; NIST SP 800-53 Rev. 5 PT family; NIST SP 800-122 (PII)

2.10.1 Personal Data Identification and Minimization

[R.AI-10.1] PERSONAL DATA IDENTIFICATION AND MINIMIZATION

The AI system shall identify personal data within all training, validation, test, and operational datasets, and shall apply data minimization principles to limit personal data collection, retention, and processing to what is necessary for the declared purpose.

Rationale: Personal data identification is a prerequisite for any privacy control. Without knowing what personal data is present, no other privacy requirement can be satisfied. Direct identifiers (name, email, government ID, biometric data, medical record number) are typically straightforward to detect; indirect identifiers (combinations of attributes that enable re-identification) require more sophisticated assessment. Data minimization processing only what is necessary for the declared purpose is foundational to GDPR Article 5(1)(c), CCPA reasonable necessity provisions, and HIPAA minimum necessary standard. AI systems often violate minimization implicitly by retaining all collected data for potential future training; explicit minimization requires documented justification for each data category retained.

[V.AI-10.1] PERSONAL DATA IDENTIFICATION AND MINIMIZATION

The AI system shall verify personal data identification and minimization by inspection and analysis.

The inspection shall confirm that personal data identification methods are documented and applied to all datasets in scope; that direct and indirect identifiers are documented per dataset; that data minimization analysis documents the necessity of each personal data category retained; and that minimization decisions are reviewed at the cadence of the dataset refresh cycle.

The analysis shall confirm that automated identification tools are validated against known personal data samples; that re-identification risk assessment addresses indirect identifier combinations; and that retention periods for personal data are bounded and justified relative to the declared processing purpose.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that personal data identification methods are documented and applied, identifiers are documented per dataset, minimization analysis is complete with documented necessity, automated tools are validated, re-identification risk is assessed for indirect identifiers, and retention periods are bounded and justified.

2.10.2 Lawful Basis and Consent Management

[R.AI-10.2] LAWFUL BASIS AND CONSENT MANAGEMENT

The AI system shall document the lawful basis for personal data processing under each applicable privacy framework, and shall implement consent management mechanisms where consent is the lawful basis.

***Rationale:** Personal data processing requires a lawful basis under all major privacy frameworks. GDPR Article 6 enumerates six lawful bases (consent, contract, legal obligation, vital interests, public task, legitimate interests); HIPAA distinguishes covered entity processing, business associate processing, and authorization-based processing; CCPA provides a notice-and-opt-out framework with specific bases for sensitive personal information. The lawful basis selected determines the obligations that follow consent-based processing requires consent records, withdrawal mechanisms, and re-consent triggers; legitimate-interests-based processing requires a documented balancing test. AI systems trained on data collected under one lawful basis may be unable to be retrained on the same data if the basis changes; documenting basis throughout the lifecycle is necessary for ongoing lawful operation.*

[V.AI-10.2] LAWFUL BASIS AND CONSENT MANAGEMENT

The AI system shall verify lawful basis and consent management by inspection.

The inspection shall confirm that the lawful basis for personal data processing is documented under each applicable privacy framework; that the basis is documented per dataset and per processing purpose; that consent records are maintained where consent is the basis, including consent text, capture mechanism, timestamp, and withdrawal status; that consent withdrawal mechanisms are implemented and accessible to data subjects; and that processing ceases or is justified under a different basis when consent is withdrawn.

Success Criteria: The verification shall be considered successful when the inspection shows that lawful basis is documented per dataset and processing purpose under each applicable framework, consent records are complete and

accessible, withdrawal mechanisms are functional, and basis changes trigger documented review.

2.10.3 Data Subject Rights Implementation

[R.AI-10.3] DATA SUBJECT RIGHTS IMPLEMENTATION

The AI system shall implement procedures to support data subject rights as required by applicable privacy frameworks, including rights of access, rectification, erasure, restriction, portability, and objection.

Rationale: *Data subject rights are a defining feature of modern privacy frameworks. GDPR Articles 15 through 22 establish access, rectification, erasure, restriction, portability, objection, and rights regarding automated decision-making. HIPAA establishes patient access rights and amendment rights under 45 CFR 164.524 and 164.526. CCPA establishes access, deletion, and opt-out rights. AI systems present unique implementation challenges: the right to erasure may require model retraining if personal data influenced learned parameters; the right to access may require explanation of automated decisions; the right to rectification of training data does not retroactively correct already-trained model behavior. These challenges require explicit procedural design rather than ad-hoc handling. Where data subject rights require explanation of automated decisions, AI-8 (Explainability) outputs serve as supporting evidence.*

[V.AI-10.3] DATA SUBJECT RIGHTS IMPLEMENTATION

The AI system shall verify data subject rights implementation by inspection and test.

The inspection shall confirm that procedures are documented for each data subject right applicable under the relevant privacy frameworks; that response timelines meet regulatory requirements (e.g., 30 days under GDPR Article 12(3), 30 days under HIPAA 164.524, 45 days under CCPA); that procedures address AI-specific implementation challenges including erasure from trained models and access to automated decision explanations; and that data subject request intake mechanisms are accessible and documented.

The test shall confirm that representative data subject rights requests can be processed end-to-end within required timelines; that erasure requests result in documented removal from datasets and assessment of model retraining necessity; and that access requests produce intelligible responses including, where applicable, explanation of automated decision logic per AI-8.

Success Criteria: The verification shall be considered successful when the inspection and test show that procedures exist for each applicable right, timelines

meet regulatory requirements, AI-specific challenges are addressed procedurally, intake mechanisms are accessible, end-to-end processing meets timelines, erasure assessments include model retraining necessity, and access responses include automated decision explanation where applicable.

2.10.4 Privacy-Enhancing Techniques

[R.AI-10.4] PRIVACY-ENHANCING TECHNIQUES

The AI system shall apply privacy-enhancing techniques appropriate to the personal data sensitivity, processing purpose, and re-identification risk of the system.

Rationale: *Privacy-enhancing techniques (PETs) reduce privacy risk through technical design rather than procedural controls alone. Anonymization removes the link between data and data subjects; pseudonymization replaces identifiers with reversible tokens; differential privacy adds calibrated noise to prevent membership inference; federated learning trains models without centralizing raw data; secure aggregation enables computation without exposing inputs; encryption protects data at rest, in transit, and increasingly in computation. The selection of PETs depends on the threat model, the utility requirements, and the regulatory regime. PETs are not interchangeable anonymization that meets HIPAA Safe Harbor may not meet GDPR anonymization standards; differential privacy parameters that satisfy academic definitions may not satisfy regulatory expectations. Where differential privacy is applied as defense against model inversion attacks, see also AI-7.3 (Model Inversion and Extraction Protection); AI-10.4 addresses PETs as design choices for data subject privacy, while AI-7.3 addresses inversion as an adversarial threat. The same technical mechanism may satisfy both requirements.*

[V.AI-10.4] PRIVACY-ENHANCING TECHNIQUES

The AI system shall verify privacy-enhancing techniques by analysis and test.

The analysis shall confirm that the privacy-enhancing techniques applied are appropriate to the personal data sensitivity, processing purpose, and re-identification risk; that the selection rationale is documented including consideration of alternatives; that anonymization or pseudonymization claims are validated against the applicable regulatory standard (e.g., GDPR Article 4(5), HIPAA Safe Harbor at 45 CFR 164.514(b)(2), or Expert Determination at 164.514(b)(1)); that differential privacy parameters where applied are documented with privacy budget and composition analysis; and that encryption methods meet applicable cryptographic standards.

The test shall confirm that applied techniques function as designed under representative operational conditions; that anonymization is robust against re-identification using documented adversary capability assumptions; and that privacy guarantees do not degrade unacceptably under operational use patterns.

Success Criteria: The verification shall be considered successful when the analysis and test show that PETs are appropriate to risk and purpose, selection is documented with alternatives considered, anonymization meets the applicable regulatory standard, differential privacy parameters are documented with composition analysis, encryption meets cryptographic standards, techniques function under operational conditions, and privacy guarantees are robust against documented adversary assumptions.

2.10.5 Cross-Border Data Transfer Controls

[R.AI-10.5] CROSS-BORDER DATA TRANSFER CONTROLS

The AI system shall identify and control cross-border transfers of personal data using mechanisms appropriate to the source and destination jurisdictions.

Rationale: *Personal data transferred across jurisdictional boundaries is subject to additional regulatory requirements. GDPR Chapter V restricts transfers outside the European Economic Area to jurisdictions with adequacy decisions, transfers under Standard Contractual Clauses, transfers under Binding Corporate Rules, or transfers under enumerated derogations. Other jurisdictions impose analogous restrictions (e.g., China PIPL, India DPDP, Brazil LGPD). AI systems often involve transfers that are not obvious cloud-hosted training infrastructure, third-party annotation services, regional inference endpoints, model weights derived from trans-jurisdictional training data. Each transfer point must be identified and brought under an appropriate transfer mechanism, or the system must be architected to avoid the transfer.*

[V.AI-10.5] CROSS-BORDER DATA TRANSFER CONTROLS

The AI system shall verify cross-border data transfer controls by inspection and analysis.

The inspection shall confirm that personal data flows are mapped across all AI lifecycle stages including data collection, training infrastructure, third-party services, model storage, and inference endpoints; that cross-border transfers are identified with source and destination jurisdictions; that transfer mechanisms are documented for each transfer (adequacy decision, SCCs, BCRs, derogations, or other applicable mechanism); and that jurisdictional limitations of system

deployment are documented in the ODD under Regulatory and Operational Authorization.

The analysis shall confirm that selected transfer mechanisms are valid under current regulatory guidance; that transfer impact assessments have been performed where required (e.g., post-Schrems II requirements for transfers to certain jurisdictions); and that contingency plans exist for transfer mechanism invalidation.

Success Criteria: The verification shall be considered successful when the inspection and analysis show that personal data flows are mapped, cross-border transfers are identified with jurisdictions, transfer mechanisms are documented and valid, transfer impact assessments are complete where required, and contingency plans exist for mechanism invalidation.

2.10.6 Privacy Impact Assessment

[R.AI-10.6] PRIVACY IMPACT ASSESSMENT

The AI system shall conduct a Privacy Impact Assessment (PIA) or Data Protection Impact Assessment (DPIA) where required by applicable regulation addressing privacy risks across the AI lifecycle and documenting risk mitigation measures.

Rationale: *Privacy impact assessment is mandated for high-risk processing under GDPR Article 35 and recommended as a best practice across most modern privacy frameworks. AI systems frequently meet GDPR DPIA triggering criteria: systematic evaluation of personal aspects (Article 35(3)(a)), large-scale processing of special categories (Article 35(3)(b)), and systematic monitoring (Article 35(3)(c)). Beyond regulatory compliance, PIAs surface privacy risks that would otherwise be invisible: re-identification risk in anonymized datasets, membership inference susceptibility, function creep beyond original purpose, and downstream privacy harms from automated decisions. The PIA is the integrating artifact that ties together AI-10's other sub-requirements.*

[V.AI-10.6] PRIVACY IMPACT ASSESSMENT

The AI system shall verify privacy impact assessment by inspection.

The inspection shall confirm that a PIA or DPIA has been conducted addressing the AI system; that the assessment covers personal data flows, lawful basis, data subject rights implementation, privacy-enhancing techniques applied, cross-border transfers, and re-identification and inference attack risks; that the assessment is traceable to the declared ODD Subject Population (AI-1.0); that mitigation measures are documented for each identified risk; that the assessment has been reviewed and approved by an authority with privacy responsibility (e.g., Data Protection

Officer where required, Privacy Officer, or designated equivalent); and that the assessment is reviewed at periodic intervals or upon material change to the system.

Success Criteria: The verification shall be considered successful when the inspection shows that the PIA is complete and covers all required topics, is traceable to the declared ODD, includes documented mitigation measures, is approved by a privacy authority, and is subject to periodic review and change-triggered review.

2.10.7 Privacy Compliance Audit Trail

[R.AI-10.7] PRIVACY COMPLIANCE AUDIT TRAIL

The AI system shall maintain an audit trail of privacy-relevant events including data subject rights requests and dispositions, consent grants and withdrawals, cross-border transfer authorizations, identified privacy incidents, and PIA reviews and updates.

Rationale: *Regulatory compliance under GDPR, HIPAA, CCPA, and analogous frameworks requires demonstrable evidence of proper privacy handling. GDPR Article 30 specifies records of processing activities; HIPAA 164.316 requires policies and procedures documentation with retention; CCPA requires verification records for data subject requests. Beyond compliance, the privacy audit trail supports incident scoping, regulatory inquiries, and demonstration of accountability. This requirement is the privacy-domain complement to the classification compliance audit trail established in AI-1.8 the two audit trails serve distinct regulatory regimes and are maintained separately even where the underlying logging infrastructure is shared.*

[V.AI-10.7] PRIVACY COMPLIANCE AUDIT TRAIL

The AI system shall verify privacy compliance audit trail by inspection.

The inspection shall confirm that data subject rights requests are logged with timestamps, requestor identification (or appropriate pseudonymous identifier), request type, response timeline, and disposition; that consent grants and withdrawals are logged with timestamps and consent text version; that cross-border transfer authorizations are logged with destination jurisdiction and transfer mechanism; that identified privacy incidents are logged with timestamps, discovery method, scope estimate, notification actions, and remediation; that PIA reviews and updates are logged with timestamps and approver; and that audit trail records are retained for periods consistent with applicable regulatory requirements.

Success Criteria: The verification shall be considered successful when the inspection shows that data subject rights, consent events, transfer authorizations,

privacy incidents, and PIA reviews are logged with required content, and that retention periods meet applicable regulatory requirements under each privacy framework that applies to the system.

Section 3: Verification

3.1 VERIFICATION METHODS

Verification of AI-specific requirements uses standard methods defined in IEEE 1012, with tailoring for AI/ML system characteristics.

Method	Definition	AI Application
Inspection	Examination of documentation, artifacts, or code	Dataset documentation, provenance records, log schemas
Analysis	Evaluation using mathematical or analytical techniques	Bias risk assessment, coverage analysis, distribution characterization
Demonstration	Observation of system behavior under realistic conditions	Operator comprehension assessment, human-AI interaction evaluation
Test	Execution under controlled conditions with pass/fail criteria	Drift detection, OOD detection, hallucination detection

3.2 VERIFICATION MATRIX SUMMARY

The verification matrix traces each sub-requirement to its verification method and required evidence artifact.

Req ID	Requirement Summary	Method	Evidence
AI-1.0	Operational Design Domain declaration	Inspection	ODD Declaration Record
AI-1.1	Dataset separation	Inspection	Partition documentation
AI-1.2	Partition use restriction	Inspection, Analysis	Access logs, partition audit
AI-1.3	Data provenance	Inspection	Provenance records

Req ID	Requirement Summary	Method	Evidence
AI-1.4	Ground truth labeling quality	Inspection, Analysis	Labeling QA report
AI-1.5	Pre-ingestion classification screening	Inspection, Analysis	Screening report, source certification
AI-1.6	Classification inheritance	Inspection, Analysis	Inheritance documentation
AI-1.7	Output classification awareness	Test, Analysis	Output marking validation
AI-1.8	Classification compliance audit trail	Inspection	Audit trail records
AI-2.1	Bias-free baseline performance	Analysis, Test	Validation report
AI-2.2	Bias-free training data	Inspection	Dataset representation report
AI-2.3	Training dataset validation	Inspection, Analysis	Dataset audit report
AI-2.4	Continuous bias monitoring	Test	Bias monitoring logs
AI-2.5	Bias impact assessment	Analysis, Test	Impact assessment records
AI-2.6	Bias indicator definition	Inspection	Indicator and threshold record
AI-2.7	Bias threshold alerting	Test	Alert test report
AI-2.8	Bias threshold response	Test	Response pathway test report
AI-2.9	Bias event logging	Inspection	Bias event logs
AI-2.10	Safety-critical bias prohibition	Analysis, Test	Protective action test report
AI-2.11	Safety-critical bias escalation	Test	Escalation test report
AI-2.12	Bias edge case definition	Inspection	Edge case catalog

Req ID	Requirement Summary	Method	Evidence
AI-2.13	Bias edge case validation	Test	Edge case test report
AI-3.1	Test coverage metric definition	Inspection	Coverage metric documentation
AI-3.2	Test coverage threshold achievement	Analysis, Test	Coverage report
AI-3.3	Operational input distribution testing	Test	Distribution test report
AI-3.4	Boundary and edge case testing	Test	Boundary test report
AI-4.1	Data drift monitoring	Test	Drift detection logs
AI-4.2	Performance threshold maintenance	Analysis, Test	Performance monitoring records
AI-4.3	Model maintenance criteria	Inspection	Update procedures, version history
AI-4.4	Output validity boundaries	Test	Boundary test report
AI-4.5	Periodic model validation	Test	Validation records
AI-4.6	Operational validation without ground truth	Inspection, Analysis	Proxy validation documentation
AI-5.1	Hallucination criteria definition	Inspection	Criteria documentation
AI-5.2	Hallucination detection	Test	Detection test report
AI-5.3	Hallucination response	Test	Response test report
AI-5.4	Hallucination logging	Inspection	Hallucination event logs
AI-5.5	Independent output validation	Analysis, Test	Independent validation report
AI-6.1	Training distribution characterization	Inspection	Distribution documentation

Req ID	Requirement Summary	Method	Evidence
AI-6.2	Runtime OOD detection	Test	OOD test report
AI-6.3	OOD response	Test	OOD response test report
AI-6.4	OOD event logging	Inspection	OOD event logs
AI-7.1	Data poisoning protection	Analysis, Test	Poisoning test report
AI-7.2	Adversarial input protection	Test	Adversarial test report
AI-7.3	Model inversion and extraction protection	Inspection, Analysis	Access control and privacy report
AI-7.4	Model integrity	Inspection	Integrity verification logs
AI-7.5	Adversarial event logging	Inspection	Security event logs
AI-7.6	AI supply chain security	Inspection	Supply chain risk assessment
AI-8.1	Explainability requirements definition	Inspection	Explainability requirements
AI-8.2	Decision explanation generation	Test	Explanation usability assessment
AI-8.3	Confidence indication	Test	Confidence calibration report
AI-8.4	Reasoning inspection capability	Test	Inspection capability test report
AI-8.5	Public disclosure support	Inspection	Disclosure artifacts and release review records
AI-9.1	Human decision authority	Test	Authority test report
AI-9.2	Operational situational awareness	Test	SA test report
AI-9.3	Trust calibration	Analysis, Test	Human factors assessment

Req ID	Requirement Summary	Method	Evidence
AI-9.4	Graceful degradation	Test	Degraded mode test report
AI-9.5	Human-AI interaction logging	Inspection	Interaction logs
AI-9.6	Operational role verification	Inspection, Analysis	Periodic review records
AI-9.7	Operator qualification	Inspection, Test	Qualification records; role-assignment controls
AI-9.8	Workload management	Analysis, Test	Workload parameter analysis and scenario test results
AI-9.9	Training program requirements	Inspection	Training curricula, records, and update procedures
AI-10.1	Personal data identification and minimization	Inspection, Analysis	PII identification and minimization record
AI-10.2	Lawful basis and consent management	Inspection	Lawful basis and consent records
AI-10.3	Data subject rights implementation	Inspection, Test	DSR procedures and response evidence
AI-10.4	Privacy-enhancing techniques	Analysis, Test	PET selection and validation report
AI-10.5	Cross-border data transfer controls	Inspection, Analysis	Data flow map, transfer mechanism documentation
AI-10.6	Privacy Impact Assessment	Inspection	PIA/DPIA record
AI-10.7	Privacy compliance audit trail	Inspection	Privacy event audit trail

3.3 DEPLOYMENT FORMAT VALIDATION

When AI models are transformed for deployment on edge compute systems, the deployment-format model shall be revalidated against the original trained model. Common transformations that require revalidation include:

- **Quantization:** Reducing model precision (e.g., FP32 to INT8) for memory and compute efficiency

- **Pruning:** Removing model weights or layers to reduce size
- **Knowledge distillation:** Training a smaller model to mimic the original
- **Format conversion:** Converting between frameworks (e.g., PyTorch to ONNX to TensorRT)

Revalidation shall demonstrate that the deployment-format model meets all performance thresholds established for the original model. Performance degradation due to transformation shall be documented and accepted within defined margins, or the transformation shall be revised. Verification evidence shall include side-by-side performance comparisons on representative test datasets. Deployment format validation applies in concert with the model maintenance criteria established under AI-4.3 and the model integrity controls established under AI-7.4; a deployed transformation shall satisfy all three control points before operational use.

Section 4: Implementation Guidance

4.1 THRESHOLD SELECTION GUIDANCE

This section provides informative guidance for threshold selection. It does not establish normative requirements. Projects shall document threshold selection rationale as part of verification evidence.

Section 2 requirements specify that projects shall define and document thresholds for bias metrics, drift detection, performance monitoring, and other AI-specific measures. This section provides example threshold ranges and selection considerations to assist projects in establishing appropriate values. These examples represent typical industry practice and should be tailored based on mission risk tolerance, operational context, and system classification.

4.1.1 Example Threshold Ranges by Requirement

Req	Metric	Example	Selection Considerations
AI-1	Demographic parity difference	≤2% for safety-critical; ≤5% for mission-critical	Tighter thresholds for crew health/safety functions; consider protected class sample sizes
AI-1	Equalized odds difference	≤3% TPR/FPR difference across groups	Critical for detection/diagnostic systems where false negatives have safety implications

Req	Metric	Example	Selection Considerations
AI-2	Data drift (PSI)	<0.1 negligible; 0.1–0.25 moderate; >0.25 significant	Alert at moderate; halt autonomous operation at significant for safety-critical
AI-2	Data drift (KL divergence)	>0.1 warrants investigation; >0.2 requires action	Monitor per feature; aggregate thresholds may mask localized drift
AI-2	Performance degradation	≤5% accuracy drop from baseline	Baseline established during V&V; consider mission phase sensitivity
AI-4	Hallucination detection rate	≥95% detection with ≤1% false positive rate	Balance detection sensitivity against operational false alarm burden
AI-5	Neuron activation coverage	≥95% safety-critical; ≥90% mission-critical	Complement with input space coverage; architecture-dependent
AI-6	OOD detection rate	≥97% detection with ≤2% false positive rate	Validate against realistic OOD scenarios from operational environment
AI-7	Confidence calibration error	ECE ≤0.05; reliability slope 0.9–1.1	Critical for operator trust calibration; validate across operational scenarios
AI-7	Confidence calibration error	ECE ≤0.05; reliability slope 0.9–1.1	Critical for operator trust calibration; validate across operational scenarios
AI-8	State change notification	≤500ms; confidence drop >15% triggers alert	Align with human factors reaction time requirements
AI-8	Manual mode transition	≤2 operator actions; ≤5 seconds	Validate under time pressure; include in crew training scenarios

4.1.2 Scaling Thresholds by Classification

Threshold stringency should scale with AI system classification:

Safety-Critical AI: Most stringent thresholds. Conservative margins. Multiple independent detection mechanisms recommended. Thresholds derived from hazard analysis and safety margins.

Mission-Critical AI: Thresholds balanced against mission objectives. May accept higher false positive rates to ensure detection. Thresholds derived from mission success criteria.

Operational Support AI: Thresholds may be relaxed where operational impact is limited. Focus on trend detection over absolute thresholds. Thresholds derived from operational efficiency considerations.

4.1.3 Threshold Documentation

When documenting threshold selection rationale, projects should capture:

- Selected threshold value and units
- Derivation basis (analysis, heritage, industry standard, simulation)
- Relationship to mission risk tolerance and safety margins
- System response when threshold is exceeded
- Validation evidence demonstrating threshold appropriateness
- Conditions under which threshold should be revisited

4.2 CONTINUOUS LEARNING REQUIREMENTS

This section provides requirements and guidance for systems employing continuous learning during operational missions. Continuous learning enables AI systems to adapt to operational environments and address drift, enhancing mission effectiveness when properly controlled.

4.2.1 Purpose

ISO/IEC 22989:2022 Section 5.11.9.2 defines continuous learning as incremental training that occurs during operation. While continuous learning can address drift and improve performance, it introduces verification challenges that require careful consideration for safety-critical systems. Continuous learning permissions shall be consistent with the ODD evolvability declaration under AI-1.0 Regulatory and Operational Authorization; learning that alters the system's operational envelope requires either approved change control (e.g., FDA PCCP) or recertification.

Rationale for Permitting Continuous Learning: Static pre-trained models face significant limitations in long-duration operations where environmental conditions cannot be fully characterized pre-deployment. Specific scenarios requiring adaptive capability include: (1) operations encountering novel phenomena or environmental conditions that preclude timely ground-based model updates; (2)

operations where terrain, lighting, and atmospheric conditions differ substantially from training data; (3) systems that must adapt to individual user variations over extended operational durations; (4) autonomous systems where target conditions may deviate from pre-deployment characterization. In these scenarios, prohibiting continuous learning would force reliance on degraded or potentially unsafe static model outputs. The enhanced controls specified below mitigate the verification challenges while preserving mission-critical adaptability.

4.2.2 Classification-Based Requirements

Safety-Critical AI: Continuous learning during operations is PERMITTED with enhanced controls. Implement: (1) Rollback capability to last verified model, (2) Approval for model updates prior to activation, (3) Shadow mode validation before operational use, (4) Automated performance bounds monitoring with automatic reversion, (5) Continuous validation per AI-4 with tightened thresholds.

Mission-Critical AI: Continuous learning permitted with controls. Implement: (1) Performance monitoring per AI-4, (2) Automated rollback at defined thresholds, (3) Logging of all model updates, (4) Periodic review of learning trajectory.

Operational Support AI: Continuous learning permitted. Implement: (1) Baseline performance tracking, (2) Anomaly alerting for significant behavior changes.

4.2.3 Catastrophic Forgetting Mitigation

Per ISO/IEC 22989:2022, continuous learning systems are susceptible to catastrophic forgetting where new learning overwrites previously acquired knowledge. Systems employing continuous learning shall document mitigation strategies, such as elastic weight consolidation, progressive neural networks, or memory replay mechanisms.

4.3 MULTI-MODEL SYSTEM CONSIDERATIONS

This section provides informative guidance for systems incorporating multiple AI models that interact or cascade. It does not establish normative requirements.

4.3.1 Purpose

Modern systems may incorporate multiple AI models operating in concert for example, a perception model feeding a planning model, or multiple specialized classifiers contributing to a fused decision. These architectures introduce failure modes not present in single-model systems.

4.3.2 Cascading Failure Considerations

When multiple AI models are chained (output of Model A becomes input to Model B), consider:

- **Error Propagation:** Errors from upstream models may be amplified or masked
- **Confidence Degradation:** Combined confidence is generally lower than individual model confidence
- **OOD Cascade:** Upstream OOD outputs may appear in-distribution to downstream models
- **Latency Accumulation:** Sequential processing adds latency affecting time-critical decisions

4.3.3 Ensemble and Voting Systems

When multiple AI models vote or contribute to a fused output, consider:

- **Correlated Failures:** Models trained on similar data may fail on the same inputs
- **Voting Threshold Selection:** Document rationale for majority, unanimous, or weighted voting schemes
- **Disagreement Handling:** Define system behavior when models disagree beyond defined thresholds

4.3.4 Confidence Aggregation and Display

When multiple AI models contribute to a decision, combined confidence shall be computed and displayed to operators using documented methods. Consider the following approaches:

Cascaded Models: For sequential model chains, combined confidence is generally the product of individual confidences (assuming independence) or computed using Bayesian propagation methods. Display both aggregate confidence and the confidence of each contributing model to enable operator diagnosis of low-confidence outputs.

Ensemble Models: For voting or fusion architectures, document whether combined confidence is computed as weighted average, minimum (most conservative), or using ensemble-specific calibration. Display the degree of model agreement alongside aggregate confidence.

Operator Display Requirements: Multi-model confidence displays shall include: (1) Combined system-level confidence clearly labeled, (2) Individual model contributions visible on demand, (3) Visual indication when models disagree beyond defined thresholds, and (4) Clear mapping between confidence levels and recommended operator actions.

4.3.5 Architecture Documentation Requirements

Multi-model AI systems should document: model interaction diagram showing data flow between AI components, interface specifications for model-to-model data exchange, combined system-level performance requirements and verification, and failure mode analysis addressing cascading and correlated failures.

4.4 NEURAL NETWORK CONSIDERATIONS

This section provides informative guidance for systems employing neural network architectures, including deep learning models, convolutional neural networks, recurrent neural networks, and transformer-based systems. It does not establish normative requirements.

4.4.1 Purpose

Neural networks represent the dominant architecture for complex AI applications in perception, prediction, and decision support. While the normative requirements in Section 2 are algorithm-agnostic, neural networks present unique implementation challenges that warrant specific guidance. These challenges stem from the fundamental nature of neural networks: behavior encoded in millions of learned parameters that cannot be meaningfully inspected, sensitivity to training data characteristics, and non-linear failure modes that differ from traditional software or simpler ML models.

4.4.2 Explainability Approaches for Neural Networks

Neural networks present inherent explainability challenges because decision logic is distributed across learned weights rather than explicit rules. Consider the following approaches based on system classification and decision criticality:

Post-Hoc Explanation Methods: Methods such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and attention visualization provide after-the-fact explanations of individual predictions. These methods are appropriate for Operational Support AI and may support Mission-Critical AI when combined with additional safeguards. Limitations include: explanations are approximations that may not reflect true model reasoning;

explanations can be manipulated by adversarial inputs; and computational overhead may preclude real-time use.

Inherently Interpretable Alternatives: For Safety-Critical AI functions, consider whether neural networks are necessary or whether interpretable alternatives (decision trees, rule-based systems, linear models) can achieve required performance. Document the rationale when neural networks are selected over interpretable alternatives for safety-critical functions.

Confidence Calibration: Neural network confidence scores (softmax outputs) are often poorly calibrated a model reporting 90% confidence may be correct only 70% of the time. Calibration techniques such as temperature scaling, Platt scaling, or isotonic regression should be applied and validated. Display calibrated confidence to operators rather than raw softmax scores.

Layer-wise Relevance and Feature Attribution: For image-based neural networks, techniques such as Grad-CAM, integrated gradients, or layer-wise relevance propagation can highlight which input regions influenced the output. These visualizations support operator verification but do not constitute proof of correct reasoning.

4.4.3 Training and Validation Considerations

Neural network training introduces risks not present in traditional software development:

Overfitting and Memorization: Neural networks with sufficient capacity can memorize training data rather than learning generalizable patterns. Indicators include: large gap between training and validation performance; high confidence on training samples with low confidence on similar validation samples; and degraded performance on distribution shifts. Mitigation includes appropriate architecture sizing, regularization techniques (dropout, weight decay), and early stopping based on validation metrics.

Training Data Requirements: Document the relationship between model architecture complexity (parameter count) and training data requirements. As a general guideline, models should have at least 10x more training samples than learnable parameters for the task, though this ratio varies by domain. Insufficient training data relative to model complexity increases overfitting risk.

Reproducibility: Neural network training involves stochastic processes (random initialization, data shuffling, dropout). Document random seeds and training configurations to enable reproducibility. For safety-critical applications, consider training multiple models with different seeds and analyzing variance in predictions.

Validation Set Contamination: Beyond the data partitioning requirements in AI-1, neural network development introduces additional contamination risks: hyperparameter tuning on validation data effectively incorporates validation information into the model; and repeated evaluation against the same test set during development causes implicit overfitting to test data. Maintain a held-out test set that is used only for final verification, not iterative development.

4.4.4 Deployment Transformation Validation

Section 3.3 establishes requirements for deployment format validation. Neural networks are particularly sensitive to the following transformations:

Quantization Sensitivity: Reducing precision (e.g., FP32 to INT8) can cause catastrophic accuracy degradation in some layers while having minimal impact on others. Layer-wise sensitivity analysis should identify critical layers that require higher precision. Post-quantization calibration using representative data can recover some lost accuracy. Validate quantized models across the full operational input distribution, not just typical cases.

Pruning Effects: Structured pruning (removing entire channels or layers) and unstructured pruning (zeroing individual weights) affect model behavior differently. Pruned models may exhibit different failure modes than original models even when aggregate metrics are similar. Validate pruned models against edge cases and adversarial inputs, not just average-case performance.

Knowledge Distillation: When training a smaller “student” model to mimic a larger “teacher” model, the student may not preserve the teacher’s behavior on out-of-distribution inputs or edge cases. Validate distilled models independently against all Section 2 requirements; inherited verification from the teacher model is insufficient.

Framework Conversion: Converting between frameworks (e.g., PyTorch to ONNX to TensorRT) can introduce numerical differences due to operator implementation variations. Validate numerical equivalence between source and target frameworks using representative inputs. Document acceptable numerical tolerance thresholds.

4.4.5 Neural Network-Specific Failure Modes

Neural networks exhibit failure modes distinct from traditional software and simpler ML models:

Adversarial Vulnerability: Neural networks are uniquely susceptible to adversarial examples inputs with imperceptible perturbations that cause dramatic misclassification. This vulnerability is fundamental to gradient-based learning and cannot be fully eliminated. Adversarial robustness testing per AI-7.2 should use techniques appropriate to the architecture: FGSM, PGD, and C&W attacks for image

models; TextFooler and character-level perturbations for text models. Defense strategies include adversarial training, input preprocessing, and certified defenses, each with tradeoffs documented.

Out-of-Distribution Behavior: Neural networks often produce high-confidence predictions on inputs far from the training distribution. Standard OOD detection (AI-6) for neural networks should consider: softmax confidence alone is insufficient OOD inputs can produce high softmax scores; ensemble disagreement, Mahalanobis distance in feature space, or dedicated OOD detection networks provide more reliable detection; and OOD detection mechanisms should be validated against realistic OOD scenarios, not just random noise.

Hallucination in Generative Models: For neural networks that generate outputs (text, trajectories, plans), hallucination detection per AI-5 requires domain-specific constraints: physical plausibility checks (conservation laws, kinematic limits); cross-validation with independent sensors or models; and temporal consistency checks for sequential outputs.

Gradient Masking: Some adversarial defenses create a false sense of security by masking gradients rather than improving robustness. Evaluate adversarial robustness using gradient-free attacks in addition to gradient-based attacks.

4.4.6 Architecture Documentation Requirements

Neural network systems should document the following in addition to standard software documentation:

Architecture Specification: Layer types, dimensions, and connectivity; activation functions and normalization schemes; total parameter count and memory footprint; and computational requirements (FLOPs) for inference.

Training Configuration: Optimizer, learning rate schedule, and regularization; batch size and training duration; data augmentation techniques; and hardware and framework versions used.

Performance Characterization: Accuracy metrics across data subsets and operational scenarios; calibration curves for confidence outputs; latency and throughput on target hardware; and known failure modes and limitations.

Deployment Artifact Chain: Traceability from trained model through all transformations to deployed artifact; validation evidence at each transformation step; and cryptographic hashes for model artifacts at each stage.

4.5 AI-SPECIFIC HAZARD ANALYSIS INTEGRATION

This section provides informative guidance for integrating AI-specific failure modes into hazard analysis. It does not establish normative requirements.

4.5.1 Purpose

Traditional hazard analysis methods (FMEA, FTA, HAZOP) assume deterministic failure modes. AI systems introduce probabilistic, emergent, and adversarial failure modes that require augmented analysis approaches.

4.5.2 AI Failure Mode Categories

When conducting hazard analysis for AI systems, include the following failure mode categories:

- **Data-Related:** bias, drift, poisoning, leakage
- **Model-Related:** hallucination, overconfidence, underfitting, overfitting
- **Operational:** OOD inputs, concept drift, degradation
- **Adversarial:** input manipulation, model theft, poisoning attacks
- **Human Factors:** automation bias, over-trust, under-trust

Appendix A: GLOSSARY

Definitions mapped to international standards where applicable.

CORE AI/ML TERMS

Artificial Intelligence (AI): Research and development of mechanisms and applications of AI systems. Computer systems (both software and hardware) designed to exhibit human-like learning, reasoning, judgement, and rational behavior. AI systems are trained (with data or knowledge, supervised, unsupervised, or reinforced) prior to deployment to approach the desired behavior. **Learned AI** indicates that learning and training occurs offline and is static in the operating environment (outcomes can still be non-deterministic). **Learning AI** indicates that continuous online learning is occurring and is dynamic in the operating environment. (ISO/IEC 22989:2022 3.1.3, tailored)

AI System: Engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives. (ISO/IEC 22989:2022 3.1.4)

Machine Learning: Process of optimizing model parameters through computational techniques, such that the model's behavior reflects the data or experience. (ISO/IEC 22989:2022 3.3.5)

Machine Learning Model: Mathematical construct that generates an inference or prediction based on input data. (ISO/IEC 22989:2022 3.3.7)

Model: Physical, mathematical or otherwise logical representation of a system, entity, phenomenon, process or data. (ISO/IEC 22989:2022 3.1.23)

Training / Model Training: Process to determine or to improve the parameters of a machine learning model, based on a machine learning algorithm, by using training data. (ISO/IEC 22989:2022 3.1.3)

Inference: Reasoning by which conclusions are derived from known premises. (ISO/IEC 22989:2022 3.1.17)

Retraining: Updating a trained model by training with different training data. (ISO/IEC 22989:2022 3.3.10)

Trained Model: Result of model training. (ISO/IEC 22989:2022 3.3.14)

Continuous Learning: Incremental training of an AI system that takes place on an ongoing basis during the operation phase of the AI system life cycle. (ISO/IEC 22989:2022 3.1.9)

Human–Machine Teaming: Integration of human interaction with machine intelligence capabilities. (ISO/IEC 22989:2022 3.3.3)

Hyperparameter: Characteristic of a machine learning algorithm that affects its learning process and is selected prior to training. (ISO/IEC 22989:2022 3.3.4)

Parameter (Model Parameter): Input variable of a model that affects how it computes its outputs (weights in a neural network). (ISO/IEC 22989:2022 3.3.8)

DATA TERMS

Training Data: Data used to train a machine learning model. (ISO/IEC 22989:2022 3.3.16)

Validation Data: Data used to compare the performance of different candidate models. (ISO/IEC 22989:2022 3.2.15)

Production Data: Data acquired during the operation phase of a deployed AI system for which a deployed system calculates a predicted output or inference. (ISO/IEC 22989:2022 3.2.12)

Test Data: Data used to assess the performance of a final model. (ISO/IEC 22989:2022 3.2.14)

Input Data: Data for which an AI system calculates a predicted output or inference. (ISO/IEC 22989:2022 3.2.9)

Ground Truth: Value of the target variable for a particular item of labelled input data. (ISO/IEC 22989:2022 3.2.7)

Label: Target variable assigned to a sample. (ISO/IEC 22989:2022 3.2.10)

RISK AND FAILURE MODE TERMS

Data Drift: Change in the statistical characteristics of production data over time compared to the training data distribution, which can degrade model prediction accuracy. (ISO/IEC 5338:2023 6.4.14 / ISO/IEC 22989:2022 5.11.9, tailored)

Concept Drift: Change in the relationship between input data and the target variable over time (i.e., the decision boundary moves), requiring relabeling of training data and model retraining. (ISO/IEC 5338:2023 6.4.14 / ISO/IEC 22989:2022 5.11.9, tailored)

Model Drift: Degradation in model performance over time due to data drift, concept drift, or other factors affecting the relationship between inputs and outputs. (Tailored)

Bias: Systematic difference in treatment of certain objects, people, or groups in comparison to others. (ISO/IEC 22989:2022 3.5.4)

Hallucination: The production of confidently stated but erroneous or false content. (NIST AI 600-1 Section 2.4)

Out-of-Distribution: Input data that falls outside the statistical distribution of the training data, for which model predictions may be unreliable or undefined. (Tailored from ISO/IEC 23053:2022)

Overfitting: Creating a model which fits the training data too precisely and fails to generalize. (ISO/IEC 23053:2022 3.2)

EXPLAINABILITY AND TRUST TERMS

Explainability: Property of an AI system to express important factors influencing the AI system results in a way that humans can understand. (ISO/IEC 22989:2022 3.5.7)

Transparency: Property of a system that appropriate information about the system is made available to relevant stakeholders. (ISO/IEC 22989:2022 3.5.15)

Predictability: Property of an AI system that enables reliable assumptions by stakeholders about the output. (ISO/IEC 22989:2022 3.5.8)

Trust Calibration: Alignment between operator trust in AI and actual AI reliability. (Human factors)

Confidence: Quantified measure of certainty or probability associated with an AI output or prediction. (Tailored)

Graceful Degradation: System capability to maintain reduced functionality when AI is impaired. (Tailored)

VERIFICATION AND VALIDATION TERMS

Verification: Confirmation, through the provision of objective evidence, that specified requirements have been fulfilled. (ISO/IEC 22989:2022 3.5.17)

Validation: Confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled. (ISO/IEC 22989:2022 3.5.18)

Continuous Validation: Ongoing monitoring process to verify that AI models continue performing satisfactorily during operation, including detection of data drift, concept drift, and performance degradation. (ISO/IEC 5338:2023 6.4.14, tailored)

Fairness Metrics: Quantitative measures used to assess whether an AI system produces equitable outputs across different groups, operational contexts, and mission scenarios. Includes metrics for disparate impact, equal opportunity, demographic parity, and performance consistency across operational conditions. (Tailored from ISO/IEC TR 24027)

SECURITY TERMS

Adversarial Attack: Deliberate manipulation of AI system inputs, training data, or model parameters intended to cause incorrect or harmful outputs. (ISO/IEC 5338:2023 6.4.3)

Adversarial Robustness: Ability of an AI system to maintain correct and safe operation despite adversarial inputs or attacks. (ISO/IEC 22989:2022 3.5.12, tailored)

Data Poisoning: Injection of malicious data into training datasets to influence model behavior in attacker-controlled ways. (ISO/IEC 5338:2023 6.4.3)

Input Manipulation: Crafting of inputs designed to cause model misclassification or incorrect outputs while appearing valid to human observers. (ISO/IEC 5338:2023 6.4.3)

Model Inversion: Attack technique that reconstructs sensitive training data by analyzing model outputs. (ISO/IEC 5338:2023 6.4.3)

Model Theft: Extraction of model parameters or behavior through systematic querying to replicate proprietary AI capabilities. (ISO/IEC 5338:2023 6.4.3)

OPERATIONAL TERMS

Fallback Behavior: Predefined safe system response invoked when AI outputs are unreliable, unavailable, or flagged as out-of-distribution. (Tailored)

Human Override: Capability for human operators to supersede or reject AI recommendations or autonomous actions, with documented rationale. (Tailored)

Autonomous Action: System action taken without requiring human approval or intervention at the time of execution. (Tailored)

Robustness: Ability of a system to maintain its level of performance under any circumstances. (ISO/IEC 22989:2022 3.5.12)

Reliability: Property of consistent intended behavior and results. (ISO/IEC 22989:2022 3.5.9)

Resilience: Ability of a system to recover operational condition quickly following an incident. (ISO/IEC 22989:2022 3.5.10)

Controllability: Property of an AI system that allows a human or another external agent to intervene in the system's functioning. (ISO/IEC 22989:2022 3.5.6)

OPERATIONAL DESIGN DOMAIN TERMS

Operational Design Domain (ODD): The operational conditions, inputs, users, subjects, regulatory context, and constraints under which an AI system is designed and validated to operate. The ODD defines the scope of the certification claim and the envelope within which reliable behavior is expected. (Framework-defined, Section 1.2.3 and AI-1.0; concept aligned with ISO 34503 and SAE J3016)

Operational Claim: Single-sentence summary statement that opens an ODD declaration, summarizing the function, subject population, operational conditions, user role, and regulatory regime of the certified system. (Framework-defined, AI-1.0)

Subject Population: The population of persons or entities that an AI system's decisions act upon, distinct from the user population that operates the system. Examples include patients in a medical decision support system, pedestrians and cyclists in a vehicle perception system, or obstacle classes in an autonomous platform. (Framework-defined, AI-1.0)

PRIVACY AND DATA PROTECTION TERMS

Personal Data: Any information relating to an identified or identifiable natural person. The precise scope varies by regulatory framework (e.g., GDPR Article 4(1), HIPAA Protected Health Information under 45 CFR 160.103, CCPA “personal information” under Cal. Civ. Code 1798.140). (Aligned with GDPR Article 4(1))

Special Categories of Personal Data: Categories of personal data subject to additional protection under GDPR Article 9 (racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data for identification, health data, sex life or sexual orientation) or analogous categories under other frameworks (e.g., HIPAA PHI; CCPA “sensitive personal information”). (Aligned with GDPR Article 9)

Data Subject: The identified or identifiable natural person to whom personal data relates. (GDPR Article 4(1))

Data Controller: The natural or legal person, public authority, agency, or other body which, alone or jointly with others, determines the purposes and means of the processing of personal data. (GDPR Article 4(7))

Data Processor: A natural or legal person, public authority, agency, or other body which processes personal data on behalf of the controller. (GDPR Article 4(8))

Lawful Basis: The legal justification for processing personal data under an applicable privacy framework (e.g., the six bases enumerated in GDPR Article 6; HIPAA permitted uses and disclosures under 45 CFR 164.502; CCPA business purposes). (Aligned with GDPR Article 6)

Anonymization: The process of rendering personal data such that the data subject is no longer identifiable, by any means reasonably likely to be used. Regulatory standards for anonymization vary (e.g., GDPR Recital 26; HIPAA Safe Harbor at 45 CFR 164.514(b)(2); HIPAA Expert Determination at 45 CFR 164.514(b)(1)). (Framework-defined; regulatory sources cited)

Pseudonymization: The processing of personal data in such a manner that the data can no longer be attributed to a specific data subject without the use of additional information, where that additional information is kept separately and subject to technical and organizational measures. (GDPR Article 4(5))

Differential Privacy: A mathematical framework for quantifying the privacy risk of a computation over a dataset, characterized by a privacy budget parameter (epsilon) that bounds the contribution of any individual record to the output. (Framework-defined; see Dwork et al. 2006 foundational literature)

Privacy-Enhancing Technique (PET): Technical method designed to reduce privacy risk through data or computation transformation, including anonymization, pseudonymization, differential privacy, federated learning, secure multi-party computation, and homomorphic encryption. (Framework-defined, AI-10.4)

Privacy Impact Assessment (PIA): Systematic assessment of the privacy risks associated with a processing activity and the mitigation measures applied. Under GDPR Article 35 and for high-risk processing, referred to as a Data Protection Impact Assessment (DPIA). (Framework-defined, AI-10.6; GDPR Article 35)

SYSTEM CLASSIFICATION TERMS

Safety-Critical AI: AI outputs directly affect human safety or system survivability. (Framework-defined, Section 1.2.2)

Mission-Critical AI: AI outputs affect operational success but not human safety. (Framework-defined, Section 1.2.2)

Operational Support AI: AI supports operations but does not drive critical decisions. (Framework-defined, Section 1.2.2)

Appendix B: AI CLASSIFICATION CHECKLIST

This checklist supports determination of AI system classification per Section 1.2.

B.1 AI PRESENCE DETERMINATION

- Does the system use behavior derived from training data rather than explicitly programmed logic?
- Are system outputs probabilistic or non-deterministic?
- Can system behavior change or degrade over time without code modifications?
- Does the system make or inform decisions affecting human safety or operational success?

Evaluation: If any answer is 'Yes' → AI is present; proceed to B.2. If all answers are 'No' → Standard software assurance applies.

B.2 SAFETY-CRITICAL AI DETERMINATION

- Do AI outputs directly affect human safety or system survivability?
- Could AI malfunction result in loss of life or critical equipment?

- Is the AI in the control path for safety-critical functions?

Evaluation: If any of 5–7 is ‘Yes’ → Safety-Critical AI. If all are ‘No’ → proceed to B.3.

B.3 MISSION-CRITICAL AI DETERMINATION

- Do AI outputs affect operational success (but not human safety)?
- Could AI malfunction result in loss of operational objectives?
- Is the AI in the decision path for mission-critical functions?

Evaluation: If any of 8–10 is ‘Yes’ → Mission-Critical AI. If all are ‘No’ → Operational Support AI.

B.4 OPERATIONAL DESIGN DOMAIN DECLARATION

Following classification determination, declare the ODD per AI-1.0. Confirm the following:

- Operational Claim summary statement drafted
- All eight ODD attribute categories addressed (Operational Environment, System Inputs, User Population, Subject Population, Operational Phases, Performance Envelope, Regulatory and Operational Authorization, Exclusions)
- Exclusions enumerated across all five sub-categories (Environments, Use Cases, Operator Conditions, Subject Conditions, System States)
- Declaration approved at the authority level specified for the system’s classification per Section 1.2.3 Table 1–4

B.5 PRIVACY APPLICABILITY DETERMINATION

- Does the system process personal data as defined under any applicable privacy framework (GDPR, HIPAA, CCPA/CPRA, PIPEDA, or analogous)?
- Does the system process special categories of personal data (e.g., GDPR Article 9, HIPAA PHI, CCPA sensitive personal information)?
- Does the system transfer personal data across jurisdictional boundaries?

Evaluation: If any answer is ‘Yes’ → AI-10 applies in full; conduct PIA/DPIA per AI-10.6. If all answers are ‘No’ → AI-10 sub-requirements may be documented as Not Applicable with rationale.

B.6 REQUIREMENT APPLICABILITY

Classification	Applicable Requirements
Safety-Critical AI	AI-1 through AI-10 (all, no tailoring without PSRB approval)
Mission-Critical AI	AI-1 through AI-6, AI-8, AI-9, AI-10 (tailoring permitted with CE approval); AI-7 tailored
Operational Support AI	AI-1 through AI-6 and AI-10 minimum; AI-7, AI-8, AI-9 as tailored

Appendix C: VERIFICATION EVIDENCE TEMPLATES

C.1 AI APPLICABILITY DETERMINATION RECORD

Field	Entry
Project Name	
System/Subsystem	
Document ID	
Date	
Prepared By	
AI Presence (B.1 Result)	☐ AI Present ☐ No AI
AI Classification (B.2/B.3)	☐ Safety-Critical ☐ Mission-Critical ☐ Operational Support
Applicable Requirements	☐ AI-1 ☐ AI-2 ☐ AI-3 ☐ AI-4 ☐ AI-5 ☐ AI-6 ☐ AI-7 ☐ AI-8
Tailoring	☐ None ☐ See attached rationale
Approved By	
Approval Date	

C.2 TEST REPORT TEMPLATE

Field	Entry
Test Report ID	
Requirement ID(s)	
Test Title	
Test Date	
Test Environment	
Test Conductor	

Field	Entry
Test Objective	
Pass/Fail Criteria	
Test Results	
Pass/Fail Determination	☐ Pass ☐ Fail
Anomalies/Observations	
Approved By	

C.3 INSPECTION RECORD TEMPLATE

Field	Entry
Inspection Record ID	
Requirement ID(s)	
Artifact Inspected	
Inspection Date	
Inspector	
Inspection Criteria	
Findings	
Compliance Determination	☐ Compliant ☐ Non-Compliant ☐ Partially Compliant
Corrective Action (if any)	

C.4 ODD DECLARATION RECORD TEMPLATE

Field	Entry
Project Name	
System/Subsystem	
Document ID	
Classification	☐ Safety-Critical ☐ Mission-Critical ☐ Operational Support
Operational Claim	<i>This AI system is designed and validated to [function] for [subject population] under [operational conditions], operated by [user role] within [regulatory regime].</i>
Operational Environment	
System Inputs (including degradation behaviors)	

Field	Entry
User Population	
Subject Population	
Operational Phases	
Performance Envelope	
Regulatory and Operational Authorization (including ODD evolvability)	
Excluded Environments	
Excluded Use Cases	
Excluded Operator Conditions	
Excluded Subject Conditions	
Excluded System States	
Approved By	
Approval Date	
Review Cadence	

C.5 PRIVACY IMPACT ASSESSMENT TEMPLATE

Field	Entry
Project Name	
System/Subsystem	
PIA ID	
Applicable Frameworks	☐ GDPR ☐ HIPAA ☐ CCPA/CPRA ☐ Other: _____
Personal Data Categories Processed	
Special Category Data	☐ Yes ☐ No (if Yes, categories: _____)
Lawful Basis (per framework)	
Data Flow Map Reference	
Cross-Border Transfers	☐ None ☐ See attached transfer documentation
Privacy-Enhancing Techniques Applied	
Re-Identification Risk Assessment	

Field	Entry
Model Inversion and Membership Inference Risk	
Data Subject Rights Implementation Summary	
Identified Privacy Risks and Mitigations	
Approving Privacy Authority	
Approval Date	
Next Review Date	

C.6 DATA SUBJECT RIGHTS REQUEST LOG TEMPLATE

Field	Entry
Request ID	
Receipt Date	
Request Type	☐ Access ☐ Rectification ☐ Erasure ☐ Restriction ☐ Portability ☐ Objection ☐ Automated Decision Explanation
Applicable Framework	☐ GDPR ☐ HIPAA ☐ CCPA/CPRA ☐ Other: -----
Requestor Verification Method	
Applicable Response Timeline	
Processing Steps	
Model Retraining Assessment (erasure only)	
Disposition	☐ Granted ☐ Partially Granted ☐ Denied (with rationale)
Response Date	
Response Delivery Method	
Approved By	

Appendix D: RISK SCORE METHODOLOGY

D.1 PURPOSE

This appendix defines the quantitative risk scoring methodology used in SCL certification assessments. The risk score is an independent output of the assessment process, issued alongside the certification determination. It quantifies assessed residual risk across four dimensions and provides a reproducible, documented measure of the system's risk posture at the time of assessment.

The risk score does not determine certification outcome. A system may receive a low risk score and still receive a Not Certified determination if requirements are not met. The score reflects assessed risk level; the determination reflects compliance status. These are independent outputs.

D.2 SCORING DIMENSIONS

The risk score is composed of four dimensions. Each dimension is scored on a 1 to 5 scale where 1 represents the lowest risk and 5 represents the highest risk.

D.2.1 Failure Mode Coverage

Assesses the extent to which AI specific failure modes have been identified, tested, and mitigated across all applicable requirement areas.

Score	Criteria
1	All applicable failure modes identified, tested, and mitigated with verified controls. No untested failure modes remain.
2	All applicable failure modes identified and tested. Mitigations verified for safety critical and mission critical failure modes. Minor gaps in mitigation verification for lower severity modes.
3	Most applicable failure modes identified and tested. Some failure modes identified but not fully tested or mitigated. Gaps are documented with risk acceptance rationale.
4	Significant failure modes identified but testing or mitigation is incomplete. Gaps exist without documented risk acceptance.
5	Failure mode identification is incomplete. Critical failure modes may be unidentified, untested, or unmitigated.

D.2.2 Consequence Severity

Assesses the operational consequence if identified failure modes occur during operations, based on the system's classification level and operational context.

Score	Criteria
1	Failure modes produce consequences limited to minor operational inconvenience. No impact on safety, mission success, or operational continuity.
2	Failure modes could degrade operational efficiency or produce incorrect advisory outputs. Human operators can detect and correct before operational impact.
3	Failure modes could affect mission success or operational objectives. Detection and correction is possible but depends on human oversight functioning as designed.
4	Failure modes could affect system survivability or create conditions requiring immediate human intervention to prevent escalation to safety impact.
5	Failure modes could directly affect human safety or create conditions where human intervention may be insufficient to prevent harm.

D.2.3 Likelihood Based on Evidence Quality

Assesses the likelihood that failure modes will occur during operations, derived from the quality, completeness, and rigor of the evidence examined during assessment.

Score	Criteria
1	Evidence is comprehensive, traceable, and independently verified. Test coverage is thorough. Continuous validation mechanisms are demonstrated and operational. Strong confidence that failure modes are controlled.
2	Evidence is complete and traceable. Testing addresses primary operational scenarios. Continuous validation is operational. Minor gaps in edge case coverage or evidence traceability.
3	Evidence addresses most requirements but gaps exist in traceability, test coverage, or validation completeness. Some reliance on process controls rather than verified technical controls.
4	Evidence is incomplete or inconsistent. Significant gaps in test coverage or validation. Controls are documented but verification is limited.
5	Evidence is absent, untraceable, or contradicted by assessment findings. Little confidence that failure modes are controlled.

D.2.4 Human Oversight Effectiveness

Assesses the effectiveness of human oversight and override controls as documented in AI-9, including the ability of operators to detect AI errors, maintain situational awareness, and exercise decision authority.

Score	Criteria
1	Human oversight is comprehensive. Decision authority is enforced through technical controls. Operators demonstrate calibrated trust, effective override capability, and graceful degradation to manual operations.
2	Human oversight is established and functional. Decision authority boundaries are defined. Minor gaps in trust calibration validation or degradation mode testing.
3	Human oversight exists but relies partially on procedural rather than technical controls. Override capability is available but has not been fully validated under operational conditions.
4	Human oversight is defined but implementation is incomplete. Operators may lack situational awareness of AI state or may not have effective override capability in all operational modes.
5	Human oversight is absent or ineffective. AI operates autonomously without functional human decision authority, or operators cannot detect AI errors in time to intervene.

D.3 SCORING PROCESS

The risk score is calculated by the Lead Assessor during Phase 3 of the assessment process. Each dimension is scored independently based on the evidence examined, findings documented, and system behavior observed during the assessment. Per dimension scores are documented with specific rationale referencing assessment findings.

The aggregation of dimension scores into the final risk score is defined in the SCL Assessment Process Documentation and is maintained separately from this framework to allow methodology refinement without framework revision.

D.4 SCORE PROPERTIES

The risk score is:

Reproducible. Another qualified assessor working from the same evidence package and assessment findings must be able to arrive at the same per dimension scores.

Scoring rationale is documented at sufficient detail to support independent verification.

Independent. The risk score does not determine the certification outcome. Requirements compliance determines the outcome. The risk score quantifies residual risk.

Point in time. The risk score reflects the assessed risk posture at the time of assessment. Operational changes, model updates, or environmental shifts after the assessment date may change the actual risk posture.

Classification aware. Consequence severity scoring inherently reflects the system’s classification level. An Operational Support system operating in an advisory capacity will score differently on consequence severity than a Safety Critical system in the control path, even if both have similar technical implementations.

D.5 SCORE REPORTING

The per dimension scores and aggregated risk score are reported in the Technical Assessment Report, the Certification Determination Document, and the Certificate (for Certified and Conditionally Certified outcomes). For Not Certified outcomes, the per dimension scores are reported in the Certification Determination Document to inform remediation prioritization.

DOCUMENT REVISION HISTORY

Version	Date	Author	Description
1.0	February 2025	Safety Critical Labs	Initial framework development
2.0	October 2025	Safety Critical Labs	Added AI-4.6 Operational Validation Without Ground Truth, AI-9.6 Operational Role Verification, Appendix D Risk Score Methodology
2.1	January 2026	Safety Critical Labs	Added AI-1.0 Operational Design Domain with Operational Claim and eight attribute categories; AI-1 renamed to Operational and Data Foundations; added Section 1.2.3 ODD procedural gate and Table 1-4 ODD Declaration Rigor; added AI-10 Privacy and Data Protection with seven sub-requirements; added Section 1.3.1 Domain Standards Citation Convention with anchor

Version	Date	Author	Description
			citations on all ten parent requirements and exemplar sub-requirement retrofits on AI-9.1, AI-8.3, AI-2.1; added cross-references for ODD, AI-10, and model integrity across Section 2; expanded Section 3.2 Verification Matrix to cover all sub-requirements; added Appendix C templates for ODD Declaration Record, Privacy Impact Assessment, and Data Subject Rights Request Log; added Appendix A glossary terms for ODD, Operational Claim, Subject Population, and privacy terminology; added Appendix B.4 ODD Declaration step and B.5 Privacy Applicability Determination; removed Appendix E Source Traceability Matrix (traceability retained inline in Appendix A); renamed the Mission Support classification tier to Operational Support throughout the framework (tier definition, tailoring authority, applicability, and worked examples).
2.1.1	April 2026	Safety Critical Labs	Added Gap 6 human factors expansion to AI-9: AI-9.7 Operator Qualification, AI-9.8 Workload Management, AI-9.9 Training Program Requirements, each with full requirement, verification, success criteria, and Applicable Domain Standards citations. Added Gap 10 public disclosure support as new AI-8.5 Public Disclosure Support (optional by classification) covering external model cards, EU AI Act Article 13 user information, EO 14110 and OMB M-24-10 federal AI inventory, and EU AI Act Article 71 public database registration. Added corresponding rows to the Section 3.2 Verification Matrix. Sub-requirement-level Applicable Domain Standards retrofit for the approximately 53 remaining sub-requirements is scheduled as a Phase 2 pass within v2.1.1.